

Options, futures & other derivatives

A complete learning and question-solving companion

Definitions that make sense. Cash flows you can follow. Examples you can calculate.

WHAT IS INSIDE

All 21 numbered sections • All 6 business snapshots • Explanations of the chapter examples and tables • Original analogies and extra examples • Solutions and methods for all 76 end-of-chapter questions



Understand the obligation first. The formula then follows naturally.

Prepared from the uploaded 9th Global Edition, ISBN 978-1-292-21289-0. Source coverage: printed pages 23-70 (PDF pages 24-71). This is an independent study companion, not an official Pearson solutions manual.

Question 2.35 requires an external historical data file that could not be retrieved. Its method, assumptions, example and reproducible code are provided; no unverified historical numerical result is claimed.

NAVIGATION

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READ THIS FIRST

How to study this guide

Your goal is to explain an unfamiliar contract by identifying who promises what, when money changes hands, and which price movement helps each party. Memorizing formulas is useful only after that story is clear.

Three passes

Pass	What to do
1 · Understand	Read the teaching pages in order. Explain each analogy aloud and reproduce the simple examples.
2 · Calculate	Attempt the textbook questions before reading the corresponding numbered solutions. Draw a cash-flow timeline for every calculation.
3 · Rehearse	Use the final oral-exam questions and formula sheet. Explain why the wrong answer is wrong.

What the source labels mean

A label such as “Hull §1.3, pp. 28-30” refers to the printed page numbers in your uploaded book. “Book example” uses the book’s data with a fresh explanation. “Extra example” is original. Exercise statements are condensed; use the source for the exact wording.

Assumptions behind the arithmetic

Unless stated otherwise, profits ignore fees, bid-ask spreads, financing costs, taxes and interest on intermediate cash flows. Payoffs are measured at expiration. When comparing cash flows at different dates rigorously, first bring them to a common date. Stock prices are assumed nonnegative in maximum-loss examples.

READ THE EDITION AS A HISTORICAL TEXT

Exchange specifications, regulations, tax rates and market statistics in this edition describe its publication-era setting. They are retained when needed to understand its exercises, not presented as current trading or tax instructions. A hedge for accounting purposes and a hedge for tax purposes are separate classifications.

Two source errors to avoid

On printed p. 53, the narrative says the gold position closes at \$1,226.90. Table 2.1 and the stated loss require \$1,426.90: $200 \times (1,426.90 - 1,450) = -\$4,620$. On p. 33, the Figure 1.3 caption says “per share,” but the plotted magnitudes describe whole 100-share contracts.

The ImportCo discussion also changes an August date in passing. Keep the contractual horizon at three months for its hedge calculations. These inconsistencies do not change the economic method.

FOUNDATIONS · HULL CHAPTER 1, PP. 23-24

1. The problem derivatives solve

Definition

A derivative is a financial instrument whose value depends on another variable, called the underlying. The underlying can be a stock price, exchange rate, interest rate, commodity price, index, default event, temperature or snowfall. It need not be a physical object.

Start with uncertainty

A Moroccan importer must pay \$100,000 in three months. At 10 MAD per dollar, that is 1,000,000 MAD. At 11, it becomes 1,100,000 MAD; at 9, only 900,000 MAD. The company is good at selling computers, but its profit also depends on a currency price it cannot control.

A forward can fix the future exchange rate. An option can cap the exchange rate the company must pay while allowing it to benefit if the dollar becomes cheaper. The contract changes the distribution of business outcomes.

Term	Plain meaning
Spot price	The price for almost immediate delivery of the underlying.
Maturity	The specified future ending or settlement date.
Counterparty	The other party to a contract.
Notional	Reference size used to determine the exposure or payments.
Payoff	The contract's value or net contractual result at a specified future date.
Profit	Payoff plus receipts minus costs, using a consistent time convention.

Analogy: a reservation right

Paying a fee for the right to buy a laptop next month for 8,000 MAD gives you something whose usefulness depends on the laptop price. If the laptop rises to 10,000, the right is valuable. If it falls to 6,000, you need not use the right. This resembles an option, with a fee paid for choice.

WHAT IS TRANSFERRED?

A derivative usually transfers exposure to a price or event. It does not necessarily transfer ownership of the underlying at the start. A zero initial contract value does not imply zero exposure.

HULL §§1.1-1.2, PP. 24-28

2. Trading venues and market size

An exchange defines standardized contracts and provides a trading venue. Standardization helps many participants trade the same instrument. OTC transactions are agreed through dealers or other arrangements and can be tailored to a company's quantity, date and other needs.

Aspect	Exchange	OTC
Terms	Listed and standardized	Often negotiable
Trading	Order matching under venue rules	Dealers, electronic venues or negotiation
Clearing	Central clearing for futures	Bilateral or central
Typical strength	Liquidity and easy offsetting	Precise fit to an exposure

A market maker quotes a bid, at which it buys, and an offer (ask), at which it sells. The customer normally buys at the offer and sells at the bid. The spread helps compensate the dealer for costs, inventory risk and liquidity provision.

From trading floors to computers

Hull traces CBOT to 1848, CME to 1919, exchange-listed calls at CBOE to 1973, and puts there to 1977. Early grain contracts solved problems of quantity, quality and future delivery. Open outcry used voices and hand signals; electronic matching enabled algorithmic and high-frequency trading. The history explains why standardization mattered before sophisticated mathematics.

Notional is not value

A contract to exchange \$100 million can have a current market value of only \$1 million. The \$100 million is a reference scale, not its purchase price or a forecast loss. Gross market value, net exposure and notional are different measures.

The book's December 2012 snapshot reports \$632.6 trillion of OTC notional, \$52.6 trillion of exchange-related underlying amounts and \$24.7 trillion of gross OTC market value. These historical measures are not directly interchangeable. Central clearing can also turn one original deal into two reported contracts.

Post-crisis direction described in this edition

The chapter describes greater use of swap execution facilities (SEFs), central counterparties (CCPs) for standardized trades, and reporting to trade repositories. A SEF helps execute trades; a CCP manages cleared obligations. These are distinct functions. Exact mandates have product and participant exemptions.

HULL §1.3, PP. 28-30

3. Forwards: an obligation for later

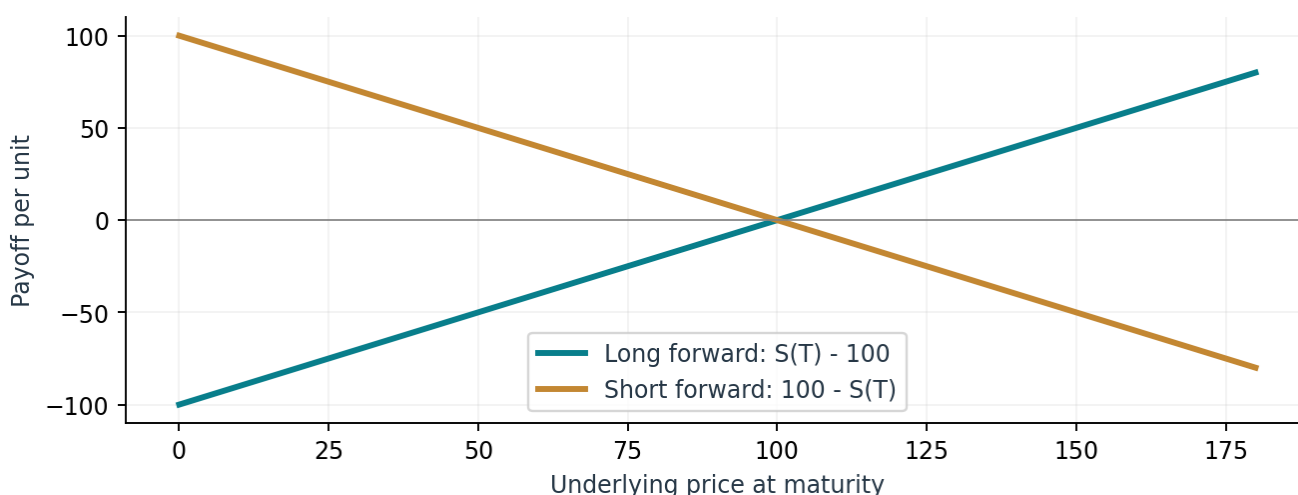
Definition: a forward is an agreement to buy or sell a specified quantity at an agreed delivery price on a future date. The long agrees to buy; the short agrees to sell. Neither party has the option to walk away merely because the market moves unfavorably.

CALCULATION

$$\text{Long payoff} = Q \times (S_T - K)$$

$$\text{Short payoff} = Q \times (K - S_T)$$

Why? The long receives an asset worth S_T and pays K . The short receives K and gives up the asset. Q converts the per-unit difference into the total result. A physically settled forward exchanges the asset and QK ; a cash-settled arrangement pays the specified difference.



At the delivery price, both payoffs are zero. Above it, the long gains and the short loses; below it, the signs reverse.

Book example: £1 million due in six months

Table 1.1 quotes the bank's six-month GBP offer at 1.5532 USD/GBP. A dollar-paying company buys pounds at the offer, locking in \$1,553,200. At $S_T = 1.6000$, the forward advantage is \$46,800. At 1.5000, it is a \$53,200 disadvantage. The dollar cost remains fixed in both cases.

SPOT, FORWARD PRICE AND CONTRACT VALUE

Spot is the underlying price now. The forward price is the delivery price available on a new fair contract. K is the fixed delivery price of your existing contract. Your contract can gain value while K stays unchanged. A fair forward usually requires no initial purchase premium, although collateral may be required.

HULL §§1.3 AND 1.7, PP. 28-35

4. Currency quotations and hedge direction

Table 1.1 quote: USD per GBP	Bank bid	Bank offer
Spot	1.5541	1.5545
One month	1.5538	1.5543
Three months	1.5533	1.5538
Six months	1.5526	1.5532

Read the units before the numbers. A rise in USD/GBP means one pound costs more dollars: the pound strengthens. The bank buys pounds at its bid and sells pounds at its offer. Do not choose the side based only on which number looks attractive.

Book example: ImportCo

ImportCo owes £10 million in three months. A stronger pound raises its dollar bill. It buys pounds forward at the three-month offer of 1.5538, fixing the bill at $10,000,000 \times 1.5538 = \text{\$15,538,000}$. At a final rate of 1.4000 the unhedged bill would be \$14 million; at 1.6000 it would be \$16 million. Hedging removes uncertainty rather than guaranteeing hindsight savings.

Book example: ExportCo

ExportCo will receive £30 million. A weaker pound reduces dollar revenue. It sells pounds forward at the three-month bid of 1.5533, fixing revenue at $30,000,000 \times 1.5533 = \text{\$46,599,000}$. Its hedge direction is opposite to ImportCo's because the business exposure is opposite.

Extra example: a Moroccan importer

If a company owes €40,000 and locks 11 MAD/EUR, its cost is 440,000 MAD. At 11.60 spot, the forward saves 24,000 relative to market; at 10.50 spot, it costs 20,000 more than market. The same certainty is purchased in both scenarios.

DIRECTION CHECK

Need to buy later and fear a rise? Long forward/futures, or buy a call. Need to sell later and fear a fall? Short forward/futures, or buy a put. State the currency you are buying or selling; "long the exchange rate" alone can be ambiguous.

HULL §§1.3-1.4 AND 1.9, PP. 30, 38-39

5. Futures and the first arbitrage argument

A futures contract provides standardized exposure to future delivery or settlement and is traded through an exchange and clearing structure. Gains and losses are paid along the way. The futures quote is a price per underlying unit or index point, not the amount you pay upfront to own the contract.

Book example: a \$60 non-dividend-paying share

With one-year borrowing and lending at 5%, carrying the share for a year costs $60 \times 1.05 = \$63$. Under the stated frictionless assumptions, that is the fair one-year forward price.

Observed forward	Trade today	Trade in one year
\$67: too high	Borrow \$60; buy the share; sell forward at \$67.	Deliver the share; receive \$67; repay \$63; keep \$4.
\$58: too low	An owner sells the share for \$60, invests proceeds, and buys forward at \$58.	Investment becomes \$63; repurchase the share for \$58; keep \$5 and restore ownership.

For a pure reverse-cash-and-carry arbitrage starting without stock, the low-price strategy requires the ability to borrow and short the share. Transaction costs, dividends, financing differences and short-sale restrictions can change the argument.

Book example: a dual-listed stock

New York price: \$150. London price: £100. Exchange rate: \$1.53/£. The London price equals \$153. Buy 100 shares in New York and sell equivalent shares in London simultaneously: gross difference = $100 \times (153 - 150) = \300 . FX conversion and settlement must also be locked in; fees can remove the opportunity.

ARBITRAGE IS NOT A FORECAST

The price difference is locked in through offsetting transactions. Simply buying something because you think its price will rise is speculation. In the textbook idealization, arbitrage requires no net investment and no possibility of loss, with a positive gain in at least some outcomes.

HULL §1.5, PP. 30-33

6. Options: the four positions

An option buyer pays a premium for a right. A call is a right to buy; a put is a right to sell. The strike K is the transaction price. The writer receives the premium and takes the corresponding obligation if exercised and assigned.

Position	Meaning	Profit per unit at expiration
Long call	Own the right to buy	$\max(S_T - K, 0) - c$
Short call	Must sell if assigned	$c - \max(S_T - K, 0)$
Long put	Own the right to sell	$\max(K - S_T, 0) - p$
Short put	Must buy if assigned	$p - \max(K - S_T, 0)$

The maximum appears because the holder can choose not to exercise. A forward lacks that choice, so its negative outcomes are not cut off. For stock options, the book generally uses 100 shares per contract: a \$3 quote costs \$300 per contract.

Exercise style

American: exercisable during the permitted period up to expiration. **European:** exercisable only at expiration. These terms do not specify geography. Selling an option you own is different from exercising it: a European option can be sold before expiration if a market exists.

Long and short do not mean bullish and bearish in every product

A long put is ownership of a put. It benefits from a fall in the underlying. A short put is an obligation to buy the underlying and is hurt by a large fall. Buying a put and selling a call can both express a bearish view, but their downside, upside and cash flows differ.

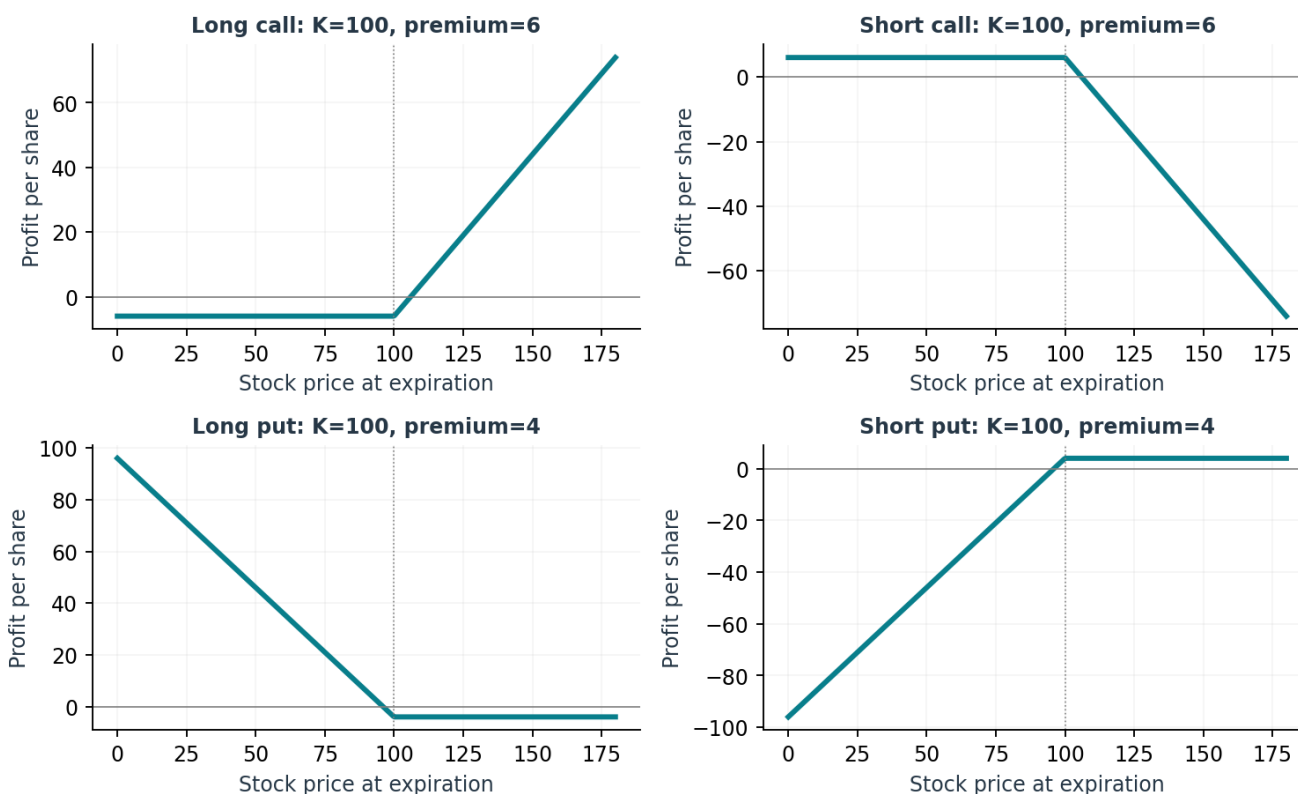
MAXIMUM LOSS

A standalone option buyer can lose the premium. An uncovered call writer has theoretically unbounded loss as the stock rises. A put writer on nonnegative stock has maximum loss $K - p$ per share. The premium limit applies to bought options, not all option strategies.

OPTIONS WORKSHOP

7. Payoff, profit and exercise

A call with $K = 100$ and premium 6 has positive exercise value when the share exceeds 100, but positive total expiration profit only above 106. A put with $K = 100$ and premium 4 has positive exercise value below 100, but positive total profit only below 96.



The flat parts come from choosing not to exercise. The premium shifts payoff into profit. Writer and buyer profits are opposites before costs.

Final price	Call payoff	Call profit	Put payoff	Put profit
90	0	-6	10	6
100	0	-6	0	-4
103	3	-3	0	-4
110	10	4	0	-4

At a final share price of 103, exercising the call recovers 3. Letting it expire recovers zero. The original premium is already spent, so exercise can be sensible even though the total trade loses money. Break-even and exercise thresholds are different.

BEFORE EXPIRATION

An option's traded value can exceed its immediate exercise value because future outcomes remain uncertain. These chapters teach expiration payoffs, not a universal rule for optimal early exercise. The moneyness test also ignores premium: in the money is not the same as profitable.

HULL TABLES 1.2-1.3, PP. 31-33

8. Reading the book's Google option tables

Tables 1.2 and 1.3 are historical bid/offer snapshots. Each row fixes a strike; columns distinguish June, September and December maturities. The underlying quote is 871.23 bid and 871.37 offer. Choose the maturity, strike, call/put and buy/sell side before reading a price.

Book example: buy a December 880 call contract

The offer is \$56.30 per share, so the 100-share contract costs **\$5,630**. If the final stock price is \$1,000, the payoff is $100 \times (1,000 - 880) = \$12,000$. Profit is $\$12,000 - \$5,630 = \mathbf{\$6,370}$. If the call expires with the stock at or below \$880, the loss is the entire premium. Expiration break-even is \$936.30.

Book example: write a September 840 put contract

The bid is \$31 per share, producing a \$3,100 receipt. At final stock price \$800, assignment requires buying 100 shares for \$84,000 when they are worth \$80,000. The \$4,000 exercise loss less the \$3,100 premium gives **-\$900**. Above \$840, the put expires and the writer keeps \$3,100. Break-even is \$809.

What changes across the table?

- Higher strike: a call's right to buy becomes less attractive, so call prices decrease; a put's right to sell becomes more attractive, so put prices increase, with other terms unchanged.
- More time: these displayed options tend to cost more. For American options, otherwise identical longer-lived exercise rights contain the shorter-lived rights. Avoid treating every European option in every setting as strictly increasing with maturity.
- Bid-ask spreads are transaction costs. In percentage terms they can be much larger for options than for liquid stock. Buy at the offer and sell at the bid in table-based exercises.

CONTRACT OR OPTION UNIT?

"100 options" in the book can mean rights on 100 shares, i.e. one standard contract. "100 option contracts" means rights on 10,000 shares when the multiplier is 100. Question 1.29 uses 100 option units, while 1.36 explicitly equates 2,000 units with 20 contracts.

HULL §§1.6-1.7, PP. 33-36

9. Hedging: combine both sides of the business

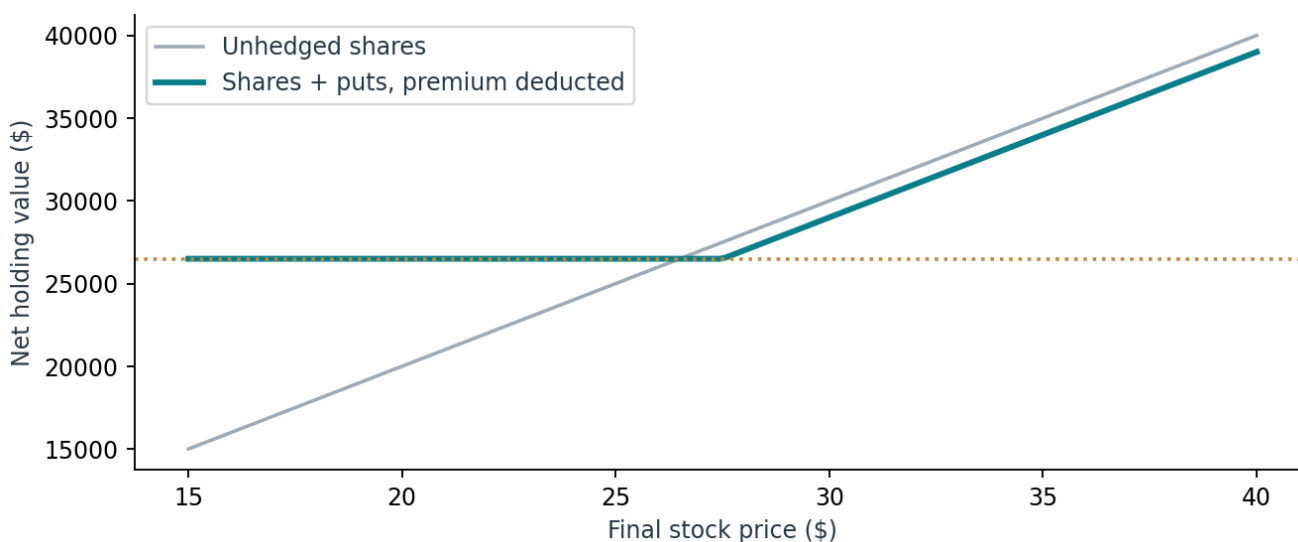
A hedger already faces a risk and uses a derivative to offset it. Assess the business plus derivative, not the derivative alone. A producer who will sell later is hurt by a fall and can sell forward. A consumer who must buy later is hurt by a rise and can buy forward.

Extra example: producer of 10,000 units

A producer sells forward at \$50. At a final spot price of \$40, physical revenue is \$400,000 and the forward gain is \$100,000. At \$60, physical revenue is \$600,000 and the forward loss is \$100,000. Both totals are \$500,000 under perfect matching.

Book example: protecting 1,000 shares

The shares are \$28 each. Ten put contracts cover all 1,000 shares, with strike \$27.50 and premium \$1 per share. Total protection cost is \$1,000. Terminal share-plus-put value is $1,000 \times \max(S_T, 27.50)$. After deducting the premium, the floor is **\$26,500**. Relative to the original \$28,000 share holding, the maximum loss including insurance is \$1,500.



The put establishes a floor, while the investor retains gains from sufficiently higher stock prices.

FORWARD VERSUS OPTION HEDGE

A matching forward fixes the transaction price and removes both favorable and unfavorable price variation. A bought option limits the adverse outcome while preserving favorable movements, at the cost of a premium. Neither eliminates every business risk: quantity, timing, basis, credit and funding may remain.

HULL §1.8, PP. 36-38

10. Speculation and leverage

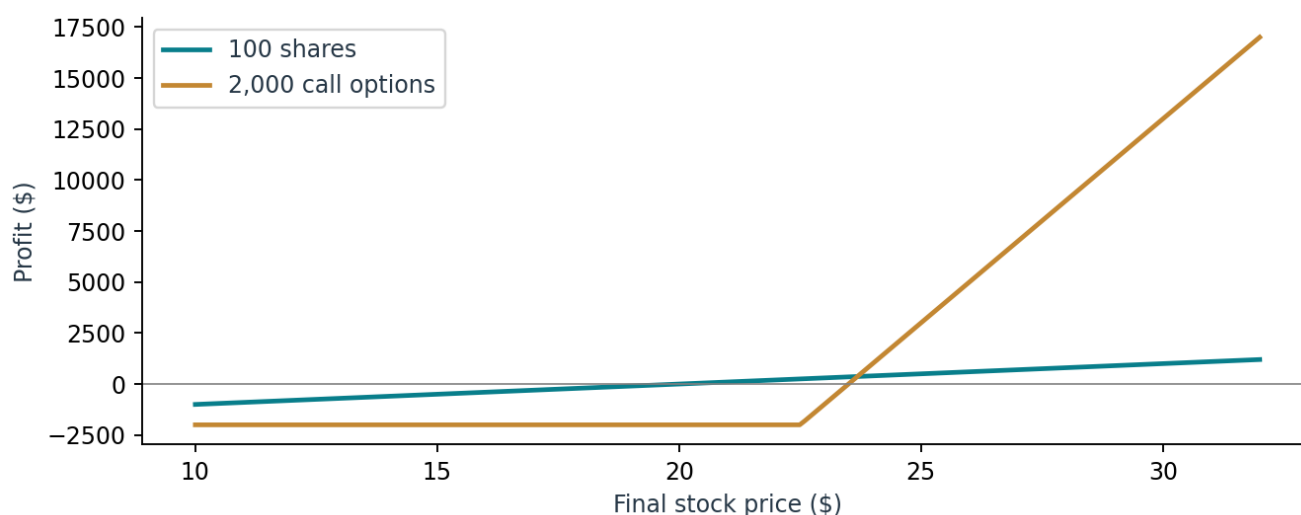
A speculator deliberately takes exposure in anticipation of a market move. Leverage means obtaining a relatively large exposure for a relatively small initial commitment. It amplifies percentage outcomes measured against that commitment.

Book example: sterling spot versus futures

The exposure is £250,000. Buying pounds at 1.5470 costs \$386,750. Four £62,500 futures at 1.5410 require the assumed \$20,000 total margin. At final rate 1.6000, spot profit is \$13,250 and futures P&L is \$14,750. At 1.5000, they are -\$11,750 and -\$10,250. The apparent difference omits interest earned or paid; it is not evidence of a free advantage from futures.

Book example: \$2,000 in stock or calls

At stock price \$20, buy 100 shares. Alternatively, buy 2,000 call units at premium \$1 and strike \$22.50. At final stock price \$27: stock profit = \$700; option profit = $2,000 \times 4.50 - 2,000 = \$7,000$. At \$15: stock loses \$500; options lose \$2,000.



The option position has a steeper upside, but can lose all invested capital even when the stock only moves modestly.

DO NOT CONFUSE THE TWO CASH COMMITMENTS

An option premium purchases a right. Futures margin supports performance and can be replenished repeatedly. A futures trader can lose more than the initial margin. A bought option's loss is limited to its premium, but the investment can still lose 100%.

HULL SNAPSHOTS 1.1-1.4, PP. 26-40

11. Business snapshots and the risk lessons

1.1 • Lehman: solvency and liquidity can interact

Hull describes Lehman's September 2008 failure through high leverage, risky mortgage exposures, short-term funding and weak influence of risk management. At leverage of 31:1, equity is about 1/31 of assets: a little over 3% of asset loss can exhaust it in a simplified balance sheet. When lenders refuse to renew funding, a firm can fail before illiquid positions can be sold calmly. Its many derivatives counterparties and collateral arrangements made resolution difficult.

1.2 • Systemic risk: a network of promises

Systemic risk is the possibility that one failure propagates through the financial system. If A fails and B loses on claims against A, B can become unable to pay C. Forced selling can also spread stress through market prices. The lesson is to examine connections and common exposures, not only each institution in isolation.

1.3 • Hedge funds: a name is not a risk guarantee

The book describes privately offered pooled investment vehicles with flexible strategies, leverage and performance-linked fees. Its example fee pattern is 1-2% of assets plus 20% of profits, and its market-size figure is historical. A fund of funds invests in several funds. A hedge fund may hedge, speculate or pursue relative-value trades.

Strategy label	What to understand
Long/short equity	Buy perceived cheap shares and short perceived expensive shares.
Convertible arbitrage	Pair a convertible bond position with a stock hedge.
Distressed securities	Invest around financial difficulty or bankruptcy.
Emerging markets	Invest in developing-market assets.
Global macro	Trade views about economies, rates, currencies and policy.
Merger arbitrage	Trade a deal's anticipated completion; failure risk remains.

1.4 • SocGen: apparent hedges can be fictitious

The chapter describes Jérôme Kerviel's large directional index exposures hidden with fictitious offsetting trades and a €4.9 billion unwind loss. It also recalls Barings/Nick Leeson and Allied Irish/John Rusnak. The lesson is independent position verification, clear mandates, risk limits and controls. Even honest compliance with a model-based limit can leave unrecognized tail and correlation risks.

HULL §2.1 AND SNAPSHOT 2.1, PP. 46-48

12. The life of a futures trade

A futures contract comes into existence when a buyer and seller open matching positions. The book's corn example is one September contract for 5,000 bushels, at 600 cents per bushel. The reference amount is $5,000 \times \$6 = \$30,000$. One party is long; one is short. The price adjusts to match willingness to buy and sell, although every completed contract always has both sides.

Stage	What happens
Order	Client specifies product, maturity, direction, quantity and order conditions.
Execution	An order matches with the other side at an agreed price.
Clearing	Positions and obligations are recorded through clearing members.
Margin and settlement	Initial collateral is posted; price changes generate daily cash flows.
Exit	Offset with an opposite trade in the same contract, or fulfill final settlement.

Closing is the opposite trade

Long one September corn contract? Sell one September corn contract to close. Short one? Buy one to close. You do not need to find the original counterparty. Selling December against a September long leaves a calendar spread, not a fully closed position. Rolling closes an old maturity and opens a new one; realized losses do not disappear.

Snapshot 2.1 · Accidental cattle delivery

Hull explicitly describes this story as possibly apocryphal. An employee meant to close a client's one-contract long cattle position but bought another contract. The institution ended up long two contracts and had to deal with delivery far away, including holding and feeding cattle until resale. The lesson is operational: "close long" means sell, and expiry calendars must be tracked.

WHY STUDY DELIVERY IF MOST TRADES CLOSE EARLY?

The possibility of delivery helps connect futures prices to the underlying market. An obligation remains real even if most participants usually offset before it is due. To avoid delivery exposure, use the product's notice dates, not just the last trading day.

HULL §2.2, PP. 48-50

13. Designing a futures contract

A futures contract is a precise specification, not merely a commodity name and price. Ambiguity about quality would let sellers tender inferior goods and would make buyers reluctant to trade.

Specification	Meaning and reason
Asset and grade	Define acceptable quality. Hull's corn example permits grades with premiums or discounts around the standard grade. Financial assets can still have deliverable baskets.
Size	Quantity per contract. Too large excludes small exposures; too small increases per-contract trading overhead.
Location	Transport and storage costs differ. The delivery location affects economic value.
Month and dates	Define listed maturities, delivery window and last trading day.
Quotation and tick	Define price units and smallest increment. Some bond quotes use 32nds; commodity quotes may use cents.
Price limits	Specify limit-up/limit-down rules. They can constrain execution during stress, not guarantee a loss cap.
Position limits	Restrict position size to help prevent excessive market control.

Book-specific examples and how to read them

The text uses 100-ounce gold, 5,000-bushel corn, 40,000-pound cattle and \$100,000-face-value Treasury bond examples. It describes grade, maturity and delivery alternatives for certain contracts. These are edition-specific specifications; learn the role of each term rather than treating old numerical specifications as current rules.

Extra example: tick value

A hypothetical contract covers 500 units, quoted in dollars per unit with a \$0.02 tick. One tick is $500 \times 0.02 = \$10$ per contract. Three contracts gaining seven ticks earn $3 \times 7 \times 10 = \$210$.

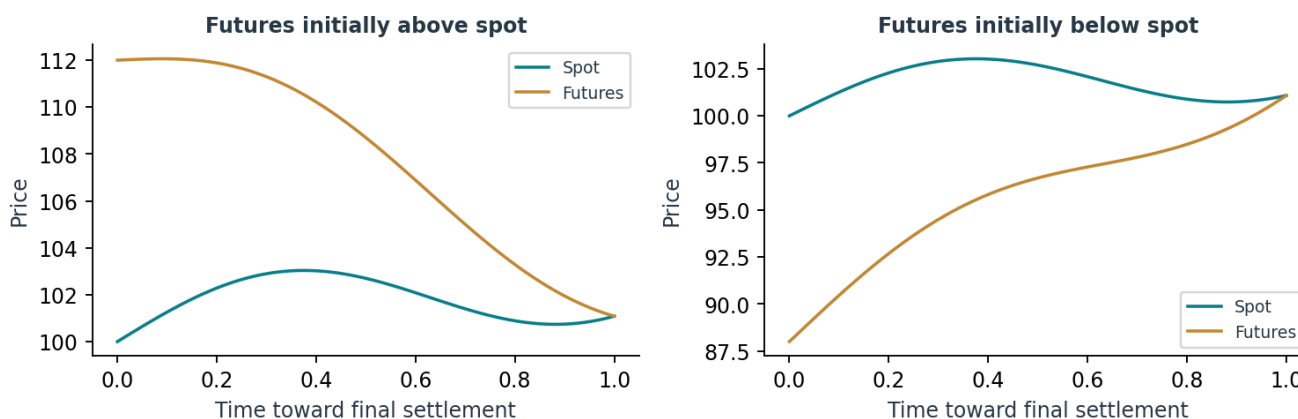
THE SHORT'S DELIVERY CHOICES HAVE VALUE

When the short can choose the cheapest eligible grade, date or location, the short has extra flexibility. Other things equal, that flexibility lowers the futures price buyers are willing to pay. The long must accept the choices permitted by the contract.

HULL §2.3, PP. 50-51

14. Why futures and spot converge

As final delivery or cash settlement approaches, the futures quote approaches the relevant spot or final reference value. For a perfectly matched asset at final settlement, $F_T = S_T$. Quality, location and timing must be aligned before comparing prices.



Illustrative paths, not forecasts. The remaining gap vanishes at final settlement in the simplified matched-contract case.

If futures is above spot during delivery

Buy the eligible asset in the spot market, short the futures, and deliver. If spot is 100 and the immediate delivery price is 105 with no relevant costs, the difference is 5. Selling pressure in futures and buying pressure in spot narrow the gap.

If futures is below spot during delivery

People who want the asset may buy futures and await delivery. This tends to raise futures demand. But a long cannot normally force immediate delivery at its preferred time: the short controls the allowed timing. Therefore the simple reverse trade is not automatically a riskless profit during the whole delivery period. Short-sale, timing and carrying constraints matter. This asymmetry is essential for Question 2.12.

Basis and curve shape

A common basis definition is $b = S - F$. Normal market: futures prices rise with maturity. Inverted market: futures prices fall with maturity. Hull's Table 2.2 also contains mixed shapes. Contango/backwardation terminology varies; do not confuse an observed curve slope with a comparison against an expected future spot price.

NOT A PURE FORECAST

A futures quote reflects trading and the economics of carrying or obtaining the asset. Financing, storage, income, immediate usefulness and risk pricing can matter. "Futures above spot" does not by itself prove that the market expects spot to rise.

HULL §2.4, PP. 51-54

15. Margin: collateral versus payment

Initial margin is the required performance collateral at entry. Maintenance margin is the customer-account threshold that triggers replenishment in the textbook model. If the account falls below maintenance, the customer tops it up to initial margin, not merely to maintenance. Failure to provide funds can lead to liquidation and a remaining debt.

CALCULATION

Long daily P&L = number of contracts $\times$ multiplier $\times$ (new settlement - previous settlement)

Short daily P&L = the negative of the long result

On entry day, use the actual trade price as the starting price. On closing day, use the exit trade price for the remaining adjustment. For a position held unchanged, the intermediate changes telescope to quantity $\times$ (exit - entry).

Extra example

Event	Price	Daily P&L	Balance before top-up	Top-up
Entry	50	-	600	600 initial
Day 1	49	-100	500	0
Day 2	48	-100	400	200
Day 3 / close	51	+300	900	0

Here quantity is 100, initial margin is 600 and maintenance is 450. On day 2, 400 is below 450, so the deposit is $600 - 400 = 200$. Total cash contributed is 800, and 900 is returned at exit: profit is 100. Deposits are funding transfers, not extra trading losses.

Daily settlement resets accumulated value

The day's loss is actually paid to the other side through clearing, rather than simply recorded as a growing unpaid debt. Immediately after settlement, the continuing futures position is economically reset at the new settlement price. The quote can be positive while the freshly settled contract value is zero.

THREE DIFFERENT THINGS

Premium purchases an option right. Initial margin supports performance. Variation settlement pays futures gains and losses. Hull also calls customer margin-call top-ups variation margin; explain whether you mean the settlement flow or the extra customer deposit.

HULL TABLE 2.1, PP. 51-53

16. The book's complete gold-margin example

Two contracts $\times$ 100 ounces = 200 ounces. Entry is \$1,450. Initial margin is \$12,000; maintenance is \$9,000. The following reconstruction uses the book's price path. Its balance column is before any new call is paid; the day-7 call is paid by day 8 and the day-11 call by day 12.

Day	Price	Daily P&L	Cumulative	Balance	Call
1	1,441.00	-1,800	-1,800	10,200	-
2	1,438.30	-540	-2,340	9,660	-
3	1,444.60	+1,260	-1,080	10,920	-
4	1,441.30	-660	-1,740	10,260	-
5	1,440.10	-240	-1,980	10,020	-
6	1,436.20	-780	-2,760	9,240	-
7	1,429.90	-1,260	-4,020	7,980	4,020
8	1,430.80	+180	-3,840	12,180	-
9	1,425.40	-1,080	-4,920	11,100	-
10	1,428.10	+540	-4,380	11,640	-
11	1,411.00	-3,420	-7,800	8,220	3,780
12	1,411.00	+0	-7,800	12,000	-
13	1,414.30	+660	-7,140	12,660	-
14	1,416.10	+360	-6,780	13,020	-
15	1,423.00	+1,380	-5,400	14,400	-
16	1,426.90	+780	-4,620	15,180	-

Day 7: $7,980 < 9,000$ triggers a call for $12,000 - 7,980 = \mathbf{4,020}$. Day 11: $8,220$ triggers a call for $\mathbf{3,780}$. Total deposits are $12,000 + 4,020 + 3,780 = 19,800$. Final balance is $15,180$. Loss = $15,180 - 19,800 = \mathbf{-4,620}$, matching $200 \times (1,426.90 - 1,450)$.

Source correction: the p. 53 prose gives 1,226.90, but the table and all calculations require 1,426.90. Excess balances on intervening days are assumed not withdrawn.

HULL §2.4, PP. 53-54

17. Clearing members and funding risk

The chain is customer, broker, clearing member, clearing house. A broker that is not a clearing member routes business through one. The clearing house computes and settles its members' obligations; customers do not simply settle directly with the strangers whose orders matched theirs.

Customer account versus clearing-member account

Customer model in this chapter	Clearing-member model in this chapter
Initial and maintenance margin; top-up after crossing the threshold.	Initial margin supports the portfolio; daily settlement losses must be paid without a maintenance buffer.
Broker may impose stricter requirements than the exchange minimum.	New positions may require additional initial margin even when existing positions made a profit.
Unused balance can sometimes be withdrawn under the account rules.	Members also contribute to a guaranty fund and meet clearing obligations.

Hull's simplified netting example has one client long 20 and another short 15, leading to margin on a net 5 contracts. Treat this as the book's example, not a universal rule allowing unrestricted netting across real customer accounts. Actual segregation and portfolio rules matter.

Collateral quality and haircuts

If an eligible security worth \$10,000 receives a 10% haircut, only \$9,000 counts toward collateral. A haircut protects against value changes and liquidation costs. Initial margin may be met with eligible securities; futures settlement losses require cash. The percentages cited for collateral in the edition are historical examples.

A hedge can work yet demand cash

A producer short futures loses cash as the commodity price rises. Its physical inventory or future sales become more valuable, but that benefit may not become cash for months. This timing mismatch is liquidity risk. A correct long-run view is not enough if interim funding fails.

CLEARING DOES NOT ERASE ALL RISK

The book's 1987 crash example shows negative customer balances and broker failures even though clearing houses honored payments to winning traders. Margin reduces accumulating credit exposure; it does not cap market losses or make the system incapable of failure.

HULL §2.5, PP. 54-57

18. OTC clearing and collateral

Bilateral clearing: the two counterparties remain responsible to each other under their contractual arrangements. A master agreement governs many trades. A credit support annex (CSA) describes collateral requirements, valuation and related terms.

Central clearing: a CCP becomes counterparty to both sides of accepted trades. Members provide initial margin, settlement or collateral flows, and default-fund contributions. Nonmembers access clearing through members.

Extra example: netting changes funding

Suppose trades against A, B and C are worth +20, -15 and -25 million to a company. With separate one-way bilateral collateral agreements, it posts $15 + 25 = 40$ million and receives nothing from A. With a single CCP, its net value is -20 million. With 10 million required initial margin, total posted funds are 30 million under these assumptions.

Variation collateral is not always final settlement

Hull distinguishes settled-to-market futures payments from OTC variation collateral securing an outstanding claim. In the book's model, cash OTC collateral earns interest; futures variation is a completed payment, not an interest-bearing loan still owed by the payer. Initial cash collateral can earn interest. Actual modern arrangements can settle some cleared OTC products to market, so read the contract.

Snapshot 2.2 · LTCM and convergence risk

LTCM bought less liquid instruments and shorted related, more liquid instruments, expecting the price gap to narrow. After Russia's 1998 default, investors placed a higher premium on liquidity. Long positions fell while shorted positions rose, creating collateral demands on both sides. Heavy leverage made it hard to wait; the book reports losses of about \$4 billion. "Convergence arbitrage" here was a risky relative-value strategy, not an unconditional riskless arbitrage.

CURRENT EXPOSURE IS NOT FUTURE EXPOSURE

A \$10 million claim with \$10 million cash collateral can have near-zero current unsecured exposure. It can still grow during default and closeout. If the claim rises to \$13 million after collateral stops arriving, there can be a \$3 million gap. Hedges may require cash even when the failed counterparty does not pay.

HULL §2.6 AND TABLE 2.2, PP. 57-59

19. Reading futures quotes and curves

Quote field	Interpretation
Open / high / low	Opening price and session extremes.
Prior settlement	Previous official settlement, used as the daily P&L baseline.
Last	Most recent transaction; not necessarily the final settlement.
Change	In Table 2.2, last minus prior settlement.
Volume	Contracts traded during the reporting period.
Contract heading	Underlying, multiplier and quotation units; essential before calculating.

Book quote: June gold

Open 1,429.5, high 1,444.9, low 1,419.7, prior settlement 1,434.3, last 1,425.3, change -9.0. Gold is 100 ounces per contract. If last becomes settlement, one long contract loses $100 \times 9 = \$900$. The reported volume of 147,943 is a trade count, not the number of ounces or net long exposure.

Other rows: use the correct units

Contract example	Quoted move	Dollar P&L on one long
June crude oil, 1,000 barrels	95.17 to 94.72 dollars/barrel	$1,000 \times (-0.45) = -450$
July corn, 5,000 bushels	655.50 to 652.50 cents/bushel	$5,000 \times (-0.03) = -150$
July soybeans, 5,000 bushels	1,419.25 to 1,418.00 cents	$5,000 \times (-0.0125) = -62.50$
July wheat, 5,000 bushels	709.75 to 710.00 cents	$5,000 \times 0.0025 = +12.50$
Cattle, 40,000 pounds	120.575 to 120.875 cents	$40,000 \times 0.003 = +120$

These calculations interpret the historical table; its last-price changes are not final daily P&L until settlement is known. Table 2.2 also has apparent year-label inconsistencies in its cattle block, so focus on the stated contract units and price changes.

The book shows upward-sloping settlement curves for some products, and mixed shapes for others. Nearby scarcity and later supply expectations can make a curve non-monotonic. A curve is a collection of prices for different contracts, not a time series of one contract.

HULL §2.6, P. 59

20. Volume and open interest

Open interest is the number of outstanding contracts. Each contract has one long side and one short side, but is counted once. Volume is the number traded over an interval. A position can change hands many times without increasing outstanding exposure.

Buyer	Seller	Change in open interest
Opens a new long	Opens a new short	+1
Closes a short	Closes a long	-1
Opens a new long	Closes an existing long	0
Closes an existing short	Opens a new short	0

Extra example: one contract, three trades

First, A opens a long against B opening a short: open interest becomes 1. Second, A sells to C, who becomes the new long: open interest remains 1. Third, C sells to B, who closes the short: open interest becomes 0. Volume was 3, even though only one contract was outstanding at a time.

A general counting shortcut

CALCULATION

Change in open interest = (number of opening sides - number of closing sides) / 2

Why divide by two? Each newly created contract contributes two opening sides. Each extinguished contract contributes two closing sides. A transfer contributes one of each and cancels out. You can also count the net change in long sides alone or short sides alone.

Book Question 2.28

Opening sides: 600 buyers + 800 sellers = 1,400. Closing sides: 1,400 buyers + 1,200 sellers = 2,600. Change in open interest = $(1,400 - 2,600)/2 = -600$. Volume is 2,000. Open interest usually declines near delivery because positions are closed or rolled to later maturities.

AN INTERVIEW TRAP

A trade requires a buyer and a seller, but not necessarily a new long and a new short. The buyer can be closing an existing short. The seller can be closing an existing long.

HULL §2.7, PP. 60-61

21. Delivery, cash settlement and dates

In the book's physical-delivery description, the short initiates a delivery notice. The clearing system assigns the obligation to an eligible long according to its rules. The original trading counterparty need not be the person who ultimately takes delivery.

Date	Meaning
First notice day	First date on which a delivery notice can be submitted.
Last notice day	Last permitted date for submitting such a notice.
Last trading day	Final opportunity to trade the contract; not necessarily the same as either notice date.

Commodity delivery often transfers a warehouse receipt. The buyer then takes responsibility for storage. Livestock can involve care and feeding. Financial delivery can use transfer of securities or currency. The invoice commonly uses the recent settlement price with any contractual grade/location adjustments.

Why the invoice is not simply the original quote

Suppose a long entered at 100 and final delivery is valued at 110. It has received cumulative futures gains of 10 per unit and pays 110 for the underlying. Net effective cost is $110 - 10 = 100$, ignoring financing. Charging only 100 at delivery while also retaining the full 10 settlement gain would double-count the benefit.

Cash settlement

A cash-settled contract ends with a final payment based on a specified reference calculation. An index is inconvenient to deliver as a full portfolio. The final reference need not be the last casually observed quote; the exchange defines it. The book describes an opening-price settlement convention for its S&P 500 example.

Extra example

A 50-dollar-per-point contract entered at 100 and finally settled at 110 has total P&L of \$500 for the long. If \$450 has already been settled, the remaining adjustment is \$50. It is not another \$500.

DELIVERY PREVENTION

A long who wants to avoid a delivery notice should exit before the relevant first-notice exposure, subject to broker deadlines. Simply waiting until the last trading day can be too late. Actual product dates must be checked before real trading.

HULL §2.8, PP. 61-62

22. Trader roles and execution orders

An FCM executes or clears customer business and charges for services; a local trades for its own account in the book's floor-trading terminology. Hedger/speculator/arbitrageur describes purpose. Scalper/day trader/position trader describes horizon: very short intervals, intraday only, or longer holding periods.

Order	Rule	What can go wrong?
Market	Execute at the best available prices promptly.	Price can differ from the last quote.
Limit	Buy at limit or below; sell at limit or above.	May never execute.
Stop	After an adverse trigger, becomes a market-type order.	A gap can cause a worse fill than the stop.
Stop-limit	After the trigger, becomes a limit order.	May remain unfilled after a gap.
Market-if-touched	After the chosen favorable price is touched, becomes market-type.	The trigger is not a guaranteed fill price.
Discretionary / market-not-held	Broker has permitted discretion over timing.	Execution depends on that discretion.

Book example: buy stop-limit

Market is 35; stop is 40 and limit is 41. When the stop condition is reached, the order becomes a buy limit at 41: buy for 41 or less if possible. If available offers jump beyond 41, it may not fill. Equal stop and limit prices are called stop-and-limit in the text.

Time conditions

Day: expires at day end. **Time-of-day:** active only during a specified interval.

Good-till-cancelled/open: remains under the venue's expiry/cancellation rules. **Fill-or-kill:** execute the full requested quantity immediately or cancel. These are distinct from the price conditions.

STOP VERSUS MARKET-IF-TOUCHED

For a long position at a current price of 35, a sell stop at 30 responds to an adverse fall. A sell MIT at 40 takes profit after a favorable rise. Both become market-type orders after triggering; a limit order retains a price restriction. Real exchanges may use protected order variants.

HULL §2.9, PP. 62-63

23. Regulation and irregularities

The chapter's US institutional framework includes the CFTC, established in 1974, and the NFA, formed in 1982. It explains public-interest supervision, reporting, registration, financial requirements, complaints, discipline and dispute resolution. Dodd-Frank expanded OTC derivatives oversight in the period discussed.

Why require reporting and financial safeguards?

A regulator cannot identify concentration or systemic stress if positions are invisible. Reporting helps reveal large exposures. Financial requirements support intermediaries' ability to honor obligations. Market rules also address fair customer treatment and abusive trading.

Cornering the market

A group accumulates large long futures positions while controlling much of the deliverable supply. Shorts approaching delivery may struggle to obtain the commodity and may buy back contracts at sharply higher prices. Hull cites the Hunt brothers' silver episode as an illustration. Responses described include higher margin, stricter position limits, restricting new speculative positions and requiring closeouts.

Other misconduct in the chapter

The text recounts a historical investigation into overcharging customers, withholding sale proceeds and front running. Front running means exploiting knowledge of a client order to trade first for oneself. It differs from legitimate market making because the trader misuses the customer information and priority.

What this teaches a software engineer

Order direction, contract identifier, maturity, multiplier, execution timestamps, independent confirmations, position limits and audit history are economically important data. A sign error can turn a closeout into a doubled position. A missing trade can hide an unhedged exposure. A stale settlement price can create the wrong margin call.

HISTORICAL CONTEXT

Learn the roles and purposes here. Exact current legal coverage, exchange rules and registration procedures are outside this edition-specific guide. The book's simplified statements about approval, limits and trading halts should not be treated as a current operating manual.

HULL §2.10, PP. 63-64

24. Accounting and tax: three clocks

The economic gain, the actual cash settlement and recognition in accounts or tax returns can occur on different schedules. These are three separate questions, even when they ultimately concern the same total gain.

Book example: a corn contract across year-end

A company enters at 650 cents, the year-end price is 670, and it closes at 680 the next February. Quantity is 5,000 bushels. Total P&L is $5,000 \times (6.80 - 6.50) = \$1,500$.

Question	Treatment in the chapter's simplified model
Cash	Paid through daily settlement as prices change.
Nonhedge accounting	Recognize \$1,000 in the first year and \$500 in the second.
Qualifying hedge accounting	Match the \$1,500 gain with the relevant hedged transaction; the example places it in the second year.

The text introduces fair-value balance-sheet recognition, disclosure, documentation and hedge-effectiveness requirements using FAS 133 and IAS 39. Those are the edition's framework. Do not automatically transfer its historical testing details to current standards. Accounting hedge qualification and tax hedge qualification are not identical.

Tax framework used for the exercises

The textbook separates ordinary income from capital gains/losses and recognizes qualifying speculative futures gains at year-end as if closed. It describes a 60% long-term / 40% short-term capital classification for the relevant noncorporate taxpayer treatment. Those percentages classify gains; they are not tax rates.

A properly identified business hedge generally produces ordinary treatment with timing matched to the hedged income or expense. For a corporate speculator, discuss annual capital recognition; do not apply an individual preferential-rate story to a corporation. The book also discusses limits and carryback/carryforward periods for losses.

HOW TO ANSWER WITHOUT INVENTING A TAX RATE

First calculate the gain in each year. Then identify hedge or nonhedge and ordinary or capital treatment under the question's stated assumptions. The question rarely provides enough to calculate a tax bill. Historical percentage rates and loss limits on p. 64 are edition-era statements, not current tax guidance.

HULL §2.11, PP. 65-66

25. Forward versus futures; FX conventions

Forward	Futures
Usually private and customizable	Exchange-standardized
Usually a specific settlement date	Often a delivery period or fixed final reference date
Basic model pays at maturity	Pays gains and losses daily
Often held to settlement	Often offset before maturity
Bilateral credit exposure unless mitigated	Clearing and margin reduce bilateral exposure

Book example: £1 million at 1.5000

Sixteen £62,500 futures provide £1 million exposure. If the final USD/GBP rate is 1.7000, both a matching forward and futures position have a nominal gain of \$200,000. The forward pays at maturity; futures pays a sequence whose algebraic sum is \$200,000. Interest on the sequence can make the economic outcomes differ.

A simple timing analogy

Receiving \$1,000 today and paying \$1,000 a year later is not the same as no cash flows: you can invest the money in between. Likewise, paying losses early and receiving offsetting gains later creates a funding cost. This is why Question 2.27 cannot be answered by the final exchange rate alone.

Currency quote direction

The book's futures quotes use USD (or US cents) per foreign-currency unit. Its spot/forward convention uses USD per GBP, EUR, AUD and NZD, but commonly the reciprocal for other major currencies. Always inspect the actual units: modern instruments can use other conventions.

CALCULATION

0.9500 USD/CAD corresponds to $1 / 0.9500 = 1.05263$ CAD/USD

A lower CAD/USD quote can mean a stronger Canadian dollar. Comparing raw numbers without conversion is meaningless. With bid/ask, invert and reverse: a 1.24/1.26 USD/EUR quote becomes 1/1.26 bid and 1/1.24 ask in EUR/USD.

PRICE VERSUS VALUE AGAIN

Futures price is a quote for the underlying/reference exposure. Contract value just after daily settlement is reset to zero in the basic model. Forward delivery price K stays fixed, while an existing forward's market value changes. These are separate fields in a valuation system.

BRIDGE TO QUESTIONS 1.22-1.41 AND 2.32

26. A method for unfamiliar payoff combinations

You do not need advanced option pricing to analyze the structured payoffs in these chapters. Write each component's terminal cash flow and add them. Divide the possible final prices at every strike; within each region the maximum functions simplify.

Identity 1: forward plus put gives a call payoff

CALCULATION

$$(S_T - K) + \max(K - S_T, 0) = \max(S_T - K, 0)$$

Below K , the put exactly cancels the forward loss. Above K , the put is zero and the forward gain remains. The combined payoff equals a call payoff. Profit still depends on premiums: a put costs p , and an actual call costs c . Equal payoff is not automatically equal profit for arbitrary initial prices.

Identity 2: call minus put gives a forward payoff

CALCULATION

$$\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$$

If S_T is above K , the call pays the difference and the put expires. Below K , the call expires and the short put loses $K - S_T$. The combination is a synthetic long forward with initial cost $c - p$. For matching European contracts, $c = p$ when K equals the fair forward price, under the usual consistent pricing assumptions.

Identity 3: a capped upside is a call spread

CALCULATION

$$\max(S_T - K_1, 0) - \max(S_T - K_2, 0), \text{ with } K_1 < K_2$$

Below the lower strike: zero. Between strikes: rising payoff. Above the higher strike: capped at $K_2 - K_1$. The same algebra explains the oil-linked note in 1.38 and the stock spread in 1.34.

RECEIPT COLLAR

For an asset you will receive, buy a put at a lower strike and sell a call at a higher strike. Terminal sale value is bounded between those strikes. Net premium must be included; the collar is not automatically free.

QUESTIONS 1.1-1.8 · SOURCE PP. 41-42

27. Chapter 1 solutions: basic positions

1.1 | Long versus short forward

The long is obligated to buy at K ; the short is obligated to sell at K . Per-unit terminal payoffs are $S_T - K$ and $K - S_T$. For the same contract, their gains and losses are equal and opposite.

1.2 | Hedge, speculate, arbitrage

A hedger reduces an existing exposure, such as a future foreign-currency bill. A speculator deliberately takes price exposure without that offsetting business need. An arbitrageur locks a discrepancy using offsetting trades. State the pre-existing exposure before classifying a trade: the instrument alone does not determine its purpose.

1.3 | Long forward at 50 versus call struck at 50

The forward obliges purchase and has payoff $S_T - 50$. The call permits purchase, has payoff $\max(S_T - 50, 0)$, and requires a premium. At $S_T = 40$, the forward loses 10; the call is not exercised, but its buyer loses the premium. At 60, both payoffs are 10, while call profit is 10 minus premium.

1.4 | Sell a call versus buy a put

Both can benefit from a falling stock price, but selling a call earns a premium and creates an obligation to sell if assigned; its uncovered loss can be unbounded. Buying a put costs a premium and creates a right to sell, with loss limited to that premium. Short-call profit is $c - \max(S_T - K, 0)$; long-put profit is $\max(K - S_T, 0) - p$.

1.5 | Short £100,000 at 1.5000 USD/GBP

Use $Q(K - S_T)$. At 1.4900: $100,000 \times 0.0100 = \text{+\$1,000}$. At 1.5200: $100,000 \times (-0.0200) = \text{-\$2,000}$. A weaker pound benefits the person who already agreed to sell it.

1.6 | Short cotton: cents must become dollars

Quantity 50,000 pounds; entry 50 cents. At 48.20 cents: $50,000 \times (0.5000 - 0.4820) = \text{+\$900}$. At 51.30 cents: $50,000 \times (0.5000 - 0.5130) = \text{-\$650}$. Forgetting the conversion creates a 100-fold error.

1.7 | Write a 40-strike put on 100 shares

You must buy 100 shares at \$40 if assigned, paying \$4,000. Let the premium per share be p . Maximum gain is $100p$; maximum loss is $4,000 - 100p$ if the stock becomes worthless. Premium is not supplied, so do not invent a numerical net profit. If European, exercise is at expiry; if American, assignment can occur earlier.

1.8 | OTC, exchange and a dealer's two prices

Exchange contracts use standardized terms and venue rules. OTC terms can be customized and clearing can be bilateral or central. A dealer's bid is the price at which it buys the quoted asset/exposure; its offer is the price at which it sells. Customers buying pay the offer, while customers selling receive the bid.

QUESTIONS 1.9-1.12 · SOURCE P. 42

28. Chapter 1 solutions: choosing an exposure

1.9 | Invest \$5,800 in shares or calls

At \$29 per share, buy 200 shares. At a \$2.90 option premium, buy 2,000 option units, or 20 standard 100-share contracts, with strike \$30. Both spend \$5,800.

CALCULATION

$$\text{Share profit} = 200(S_T - 29)$$

$$\text{Call profit} = 2,000 \max(S_T - 30, 0) - 5,800$$

Final share price	Stock profit	Option profit
20	-1,800	-5,800
29	0	-5,800
32	600	-1,800
35	1,200	4,200
40	2,200	14,200

Stock break-even is 29; call break-even is 32.90. Both investments can lose the full initial \$5,800 if stock goes to zero, but calls lose it whenever $S_T \leq 30$. For $S_T > 30$, solve $2,000(S_T - 30) - 5,800 = 200(S_T - 29)$: options outperform only above **33.3333**. Greater upside sensitivity comes with a higher hurdle.

1.10 | Insure 5,000 shares for four months

Buy puts covering 5,000 shares with a suitable strike K and expiration at or beyond the protection horizon. With 100-share contracts, this is **50 contracts**. A strike of \$25 gives gross terminal floor \$125,000 if all matching assumptions hold. Net value is reduced by premiums. Higher floors usually cost more. If expiry is later than four months, exercising at that horizon requires an appropriate exercise style, or you can sell the option.

1.11 | Does an option finance the underlying company?

A newly issued share raises funds for its issuer. A conventional exchange-traded stock option is created between market participants: the premium goes to the writer, not to the company whose stock is referenced. Company-issued warrants or compensation instruments are different arrangements; do not generalize the exchange-option answer to every company-issued security.

1.12 | Same futures, different purpose

An airline or commodity buyer can use a long futures position to reduce a pre-existing exposure to rising input costs. A trader with no such need can take the identical long position to bet on a price increase. Economic context makes one a hedge and the other speculation.

QUESTIONS 1.13-1.16 · SOURCE P. 42

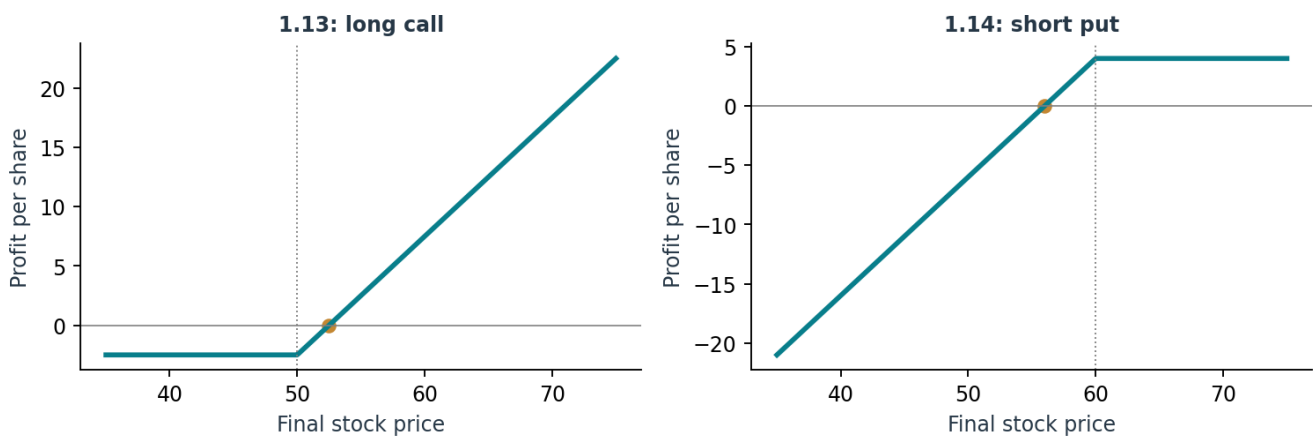
29. Chapter 1 solutions: option profit thresholds

1.13 | Long call: strike 50, premium 2.50

Profit is $\max(S_T - 50, 0) - 2.50$. It is positive when $S_T > 52.50$, zero at 52.50 and negative below. Exercise has positive value when $S_T > 50$; at exactly 50 there is no economic exercise benefit. Draw a flat line at -2.50 up to 50, then a line of slope +1 crossing zero at 52.50.

1.14 | Short put: strike 60, premium 4

Profit is $4 - \max(60 - S_T, 0)$. Below 60 it equals $S_T - 56$; at or above 60 it equals 4. The writer profits when $S_T > 56$. The holder exercises for positive value when $S_T < 60$. Thus the writer can still profit while the put is exercised, for $56 < S_T < 60$.



The kink marks the strike; the gold point marks break-even. These are not the same price.

1.15 | Write September call at 20 for 2; final stock 25

Receive **\$2 now** per share. At expiry, deliver a share for \$20 when it is worth \$25: an economic cash flow of -\$5 if the share must be purchased in the market. Total profit = $2 - 5 = \text{-\$3}$ per share. For a 100-share contract: +\$200 initially, -\$500 exercise result, net -\$300. If already covered by owned shares, delivery uses those shares, but the option's economic exercise cost remains \$5 per share.

1.16 | Write a 30-strike put for 4

Writer profit is $4 - \max(30 - S_T, 0)$. Positive gain requires $S_T > 26$; break-even is 26. Maximum gain is 4 at $S_T \geq 30$. Assignment can occur and still leave a gain when $26 < S_T < 30$.

QUESTIONS 1.17-1.22 · SOURCE P. 42

30. Chapter 1 solutions: hedges and replication

1.17 | Foreign currency to be received

Buy a put on the foreign currency, with notional equal to the receipt and a matching horizon. It gives the right to sell that currency at a minimum domestic-currency price. Buying a call would protect someone who needs to buy the foreign currency, not someone receiving it.

1.18 | US company owes CAD 1 million

(a) Buy CAD 1 million forward for six-month delivery, fixing the USD cost at $1,000,000 \times K$ when K is USD/CAD. (b) Buy calls on CAD covering CAD 1 million. At payment time, exercise if CAD is dearer than K ; otherwise buy it in the spot market. Effective USD cost before financing premium is $1,000,000 \times \min(S_T, K)$ plus the premium paid.

1.19 | Short JPY 100 million at \$0.0090/JPY

At 0.0084: $100,000,000 \times (0.0090 - 0.0084) = \text{\$60,000}$. At 0.0101: $100,000,000 \times (0.0090 - 0.0101) = \text{\$110,000}$. Preserve USD/JPY units throughout.

1.20 | Who uses Treasury bond futures?

Bond portfolio managers and dealers hedge bond-price and interest-rate exposure; speculators express views on bond prices or yields; arbitrageurs compare futures with deliverable bonds and related instruments. All else equal, higher yields mean lower bond prices. A holder fearing price falls can short suitable futures, while a future bond buyer may go long. Precise hedge sizing is a later topic.

1.21 | Why call them zero-sum?

For a given derivative, one side's contractual gain equals the other side's loss, including the premium transfer for options and using consistent cash-flow timing. Net of external transaction costs, the traders' combined result is lower. Zero-sum cash transfers do not imply zero economic usefulness: both sides can value transferring risk or obtaining a desired exposure.

1.22 | Long forward plus long put at the initial forward price

Let both strike and delivery price be K . Payoff = $S_T - K + \max(K - S_T, 0) = \max(S_T - K, 0)$. Therefore the terminal payoff is a call payoff. If put premium is p and the forward has no initial purchase cost, the simple profit is $\max(S_T - K, 0) - p$. At K equal to the fair forward price, matched European call and put values coincide under no-arbitrage assumptions. If carrying premiums to maturity, deduct p grown at the relevant rate.

HOW TO PROVE IT IN AN INTERVIEW

Split the outcomes into $S_T \geq K$ and $S_T < K$. In the first region the put is zero; in the second the put cancels the forward loss. A two-case explanation is often clearer than immediately naming a synthetic instrument.

QUESTION 1.23 · SOURCE P. 43

31. Chapter 1 solution: the currency-linked note

The ICON pays \$1,000 when $S_T \geq 169$ JPY/USD; its payoff declines below that threshold and becomes zero below 84.5. The exchange-rate quote is the opposite of the USD-per-JPY convention useful for ordinary currency calls.

Step 1 · Choose a simpler underlying variable

Define $X = 1/S_T$, measured in USD per JPY. A strengthening yen means X rises and S_T falls. The deduction before the zero floor is $\max(169,000X - 1,000, 0)$.

Step 2 · Write the floor explicitly

CALCULATION

$$\text{ICON payoff} = \max(1,000 - \max(169,000X - 1,000, 0), 0)$$

Let $a = 1/169$ and $b = 2/169 = 1/84.5$. These are USD/JPY strikes. The payoff can be written as:

CALCULATION

$$1,000 - 169,000 \max(X - a, 0) + 169,000 \max(X - b, 0)$$

Step 3 · Identify the instruments

- A zero-coupon bond paying \$1,000 at maturity.
- A short call on JPY 169,000, strike $a = 0.0059171598$ USD/JPY.
- A long call on JPY 169,000, strike $b = 0.0118343195$ USD/JPY.

Step 4 · Verify every region

Exchange-rate region	Option behavior	Total payoff
$S \geq 169$, equivalently $X \leq a$	Both calls expire.	\$1,000
$84.5 < S < 169$	Lower-strike short call pays; higher call expires.	$2,000 - 169,000/S$
$S \leq 84.5$, equivalently $X \geq b$	Both calls pay; their difference is fixed.	\$0

Check at $S = 169$: \$1,000. At $S = 100$: $2,000 - 1,690 = \$310$. At $S = 84.5$: zero. Below 84.5, the long high-strike call prevents the total payment from becoming negative.

INTUITION

The investor lends money through the bond and also sells a capped currency-option exposure. The second option is required precisely because the note's repayment has a floor at zero. Do not call the two strikes 169 and 84.5 if your option underlying is quoted in USD per JPY.

QUESTIONS 1.24-1.28 · SOURCE P. 43

32. Chapter 1 solutions: fixed payoffs and arbitrage

1.24 | Buy and later sell the same JPY amount forward

Let K_1 be the July purchase rate and K_2 the September sale rate, both USD/JPY for the same January maturity. Net payoff = $10,000,000(S_T - K_1) + 10,000,000(K_2 - S_T) = \mathbf{10,000,000(K_2 - K_1)}$ **dollars**. The final spot rate cancels. No numerical result is possible without both contracted rates. With different counterparties, legal and credit obligations remain even when price risk cancels.

1.25(a) | A very cheap 180-day sterling call

Buy the call struck at 1.42 for \$0.02 per pound and short the 180-day forward at 1.5518. Combined terminal payoff = $\max(S_T - 1.42, 0) + 1.5518 - S_T$. If $S_T \geq 1.42$, it is 0.1318. If lower, it is even greater. Thus the terminal payoff is at least **\$0.1318 per pound**, versus a \$0.02 initial outlay. Ignoring funding, the locked net gain is at least \$0.1118. If funding the premium, subtract its maturity value; the arbitrage holds whenever that value is below \$0.1318.

1.25(b) | A 90-day sterling put struck at 1.49

The 90-day forward price is 1.5556, above the strike. A \$0.02 put is not shown to violate no-arbitrage by these inputs. Long forward plus put would have terminal floor $1.49 - 1.5556 = -0.0656$ before premium: no guaranteed profit. The option's possible value depends on uncertainty and other inputs. **No arbitrage follows merely from the quoted prices.**

1.26 | Can exercising a bought call accompany a loss?

Yes. With strike 30 and premium 3, exercise has positive value for $S_T > 30$, but profit is positive only above 33. At 32, exercise produces 2, so total loss is 1. Not exercising would lose the full 3.

1.27 | Maximum gain and loss of a written option

For a put with $K = 40$ and premium 5: maximum gain **5**; maximum loss **35** per share, at zero stock price. For a call with the same strike and premium: maximum gain **5**; uncovered maximum loss is **unbounded**. Multiply by the number of underlying shares for contract totals.

1.28 | Owned stock plus a put as insurance

At maturity, stock plus put is $S_T + \max(K - S_T, 0) = \max(S_T, K)$. Thus K is the gross floor, while the upside remains. The premium is the insurance cost. This protection lasts only for the option's covered amount and period.

QUESTIONS 1.29-1.33 · SOURCE PP. 43-44

33. Chapter 1 solutions: investment alternatives

1.29 | Google shares versus 100 call units

The stock offer is 871.37; the September 880 call offer is 41.60. “100 call options” means 100 share-level option units, i.e. one standard contract, not 100 contracts.

Alternative	Upfront cost	Profit at 950	Profit at 800
100 shares	\$87,137	$100 \times (950 - 871.37) = \$7,863$	$100 \times (800 - 871.37) = -\$7,137$
100 call units	\$4,160	$100 \times (950 - 880) - 4,160 = \$2,840$	-\$4,160

The call position costs much less, so absolute profit and percentage return tell different stories. Do not compare them as equal-dollar investments.

1.30 | Dual-listed mining stock

Toronto price CAD 52 equals $52/1.01 = \text{USD } 51.48515$. Buy in New York at USD 50 and sell equivalent shares in Toronto at CAD 52, locking currency conversion. Gross gain = USD 1.48515 per share, or CAD 1.50. New York buying tends to raise its price; Toronto selling tends to lower its price. Fees and practical settlement/shorting constraints must fit inside the difference.

1.31 | Forward versus a 1,000-strike call costing 100

Forward profit is $S_T - 1,000$. Call profit is $\max(S_T - 1,000, 0) - 100$. Plot the forward as a slope-1 line crossing zero at 1,000. Plot the call flat at -100 up to 1,000, then slope 1 crossing zero at 1,100. At 800: -200 versus -100; at 1,200: +200 versus +100. The call limits downside in exchange for its premium.

1.32 | One July 40-strike put written for 3

Receive \$300 for the 100-share contract and accept the obligation to buy 100 shares for \$4,000 if assigned. Profit = $300 - 100 \max(40 - S_T, 0)$. It is positive above **37**, with maximum \$300 and maximum loss **\$3,700**. American-style assignment may occur early. A large fall creates both economic loss and the need to fund purchase.

1.33 | A bill of EUR 3 million in three months

Buy EUR 3 million forward to fix the future USD cost, using the actual three-month forward rate, not automatically the current spot 1.3500. Alternatively buy EUR calls covering the amount and horizon. Calls provide a ceiling before premium while retaining benefit if the euro weakens. No numerical forward cost or option premium is determined by spot alone.

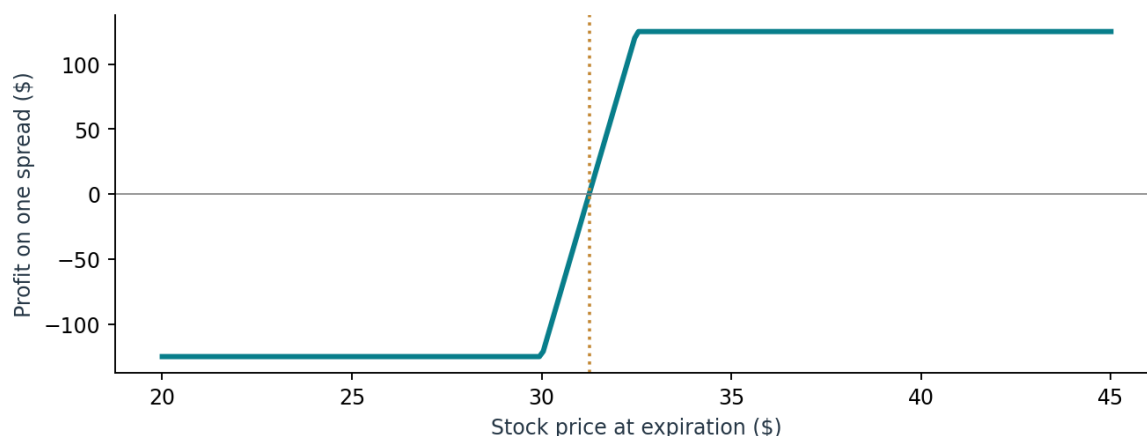
QUESTIONS 1.34-1.37 · SOURCE P. 44

34. Chapter 1 solutions: spreads and protection

1.34 | Buy the 30 call; sell the 32.50 call

This is a bull call spread. Net premium per share = $2.75 - 1.50 = 1.25$. Per-share payoff is $\max(S_T - 30, 0) - \max(S_T - 32.50, 0)$. Profit equals payoff minus 1.25.

Final-price region	Profit per share	One 100-share spread
$S \leq 30$	-1.25	-\$125 maximum loss
$30 < S < 32.50$	$S - 31.25$	Break-even at \$31.25
$S \geq 32.50$	+1.25	+\$125 maximum gain



Selling the higher-strike call lowers the premium but caps the benefit of a large rise.

1.35 | Gold cash-and-carry

Borrow \$1,400 at 4%, buy one ounce and sell it forward at \$1,500. With no storage cost or income, repay \$1,456 after one year and receive \$1,500 for delivery. Locked profit is **\$44 per ounce**. This calculation uses the exercise's annual 4% convention; another compounding convention changes the repayment slightly.

1.36 | Equal \$9,400 investment: 100 shares or 2,000 calls

Stock profit = $100(S_T - 94)$. Option profit = $2,000 \max(S_T - 95, 0) - 9,400$. Above 95, equate them: $2,000(S_T - 95) - 9,400 = 100(S_T - 94)$, giving $S_T = 100$. Calls outperform above 100. Their own break-even is 99.70. At or below 95 they lose all \$9,400; the share position need not. Choice depends on conviction, horizon and acceptable downside, not on a universal superior strategy.

1.37 | Protective put versus a stop at 820

One December 820 put contract costs $100 \times 37.50 = \$3,750$ and supplies a \$82,000 gross terminal floor for 100 shares; net of premium that is \$78,250. It retains later recovery upside while held. A sell stop has no initial option premium but can fill below 820 after a gap and can sell on a temporary fall before recovery. A stop is a trading instruction, not a guaranteed insurance payoff.

QUESTIONS 1.38-1.39 AND 1.41 · SOURCE PP. 44-45

35. Chapter 1 solutions: structured payoffs

1.38 | The oil-linked bond

Let P be the final oil price per barrel. The payment is $1,000 + 170 \max(P - 25, 0) - 170 \max(P - 40, 0)$. Therefore hold a zero-coupon bond paying \$1,000, buy calls covering 170 barrels at strike 25, and sell calls covering 170 barrels at strike 40. “170” describes underlying units; the number of exchange contracts would depend on their multiplier.

Oil price region	Maturity payment
$P \leq 25$	\$1,000
$25 < P < 40$	$\$1,000 + 170(P - 25)$
$P \geq 40$	$\$1,000 + 170(40 - 25) = \$3,550$

The long low-strike calls create the extra payment. The short high-strike calls cancel further growth after 40, imposing the \$2,550 cap on the additional amount. This is a payoff decomposition, not a claim that the bond initially costs \$1,000.

1.39 | A sterling receipt with a floor and ceiling

For the £1 million receipt, buy a put with strike 1.52 USD/GBP and sell a call with strike 1.58, both on £1 million with matching maturity. Dollar proceeds before premium are $1,000,000 \times [S_T + \max(1.52 - S_T, 0) - \max(S_T - 1.58, 0)]$.

Final USD/GBP	Gross rate obtained
Below 1.52	1.52: put supplies the shortfall.
Between 1.52 and 1.58	Actual spot rate: both options expire.
Above 1.58	1.58: written call surrenders the excess.

This is a collar. Net cost is put premium minus call premium. The requested strikes do not prove a zero-cost collar; the premiums must actually be priced.

1.41 | Buy a European call and sell the matching put

Terminal payoff = $\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$. It is a synthetic long forward. Initial cost is $c - p$. Under the standard no-arbitrage framework, $c - p$ equals the discounted difference between the fair forward price $F_{0,T}$ and K . Thus **$c = p$ when $K = F_{0,T}$** , not generally when K equals today's spot price. Income and carrying costs affect that forward price.

QUESTION 1.40 · SOURCE P. 45

36. Chapter 1 solution: pricing the currency hedges

This exercise asks for a DerivaGem calculation, introducing inputs beyond the chapter's payoff arithmetic. The values below are independently computed with the European currency-option model; DerivaGem itself was not run. Use its currency / European / Black-Scholes settings to reproduce them.

Input	Value
Spot midpoint S	$(1.5541 + 1.5545)/2 = 1.5543$ USD/GBP
Time T	$3/12 = 0.25$ years
Volatility σ	0.12 per square-root year
Domestic USD rate r	0.05, continuous-rate model input
Foreign GBP rate r_f	0.057, continuous-rate model input
ImportCo	Buy calls on £10 million with $K = 1.5700$
ExportCo	Buy puts on £30 million with $K = 1.5300$

The model, with every symbol identified

$N(z)$ is the standard normal cumulative probability. The foreign interest rate accounts for the income-like return associated with holding the foreign currency. No proof of this pricing model is required by Chapters 1-2.

CALCULATION

$$d_1 = [\ln(S/K) + (r - r_f + \sigma^2/2)T] / (\sigma\sqrt{T})$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$c = S \exp(-r_f T) N(d_1) - K \exp(-rT) N(d_2)$$

$$p = K \exp(-rT) N(-d_2) - S \exp(-r_f T) N(-d_1)$$

Results

ImportCo: $d_1 = -0.1666723$ and $d_2 = -0.2266723$. Call price = **\$0.02850682 per pound**. For £10 million, premium = **\$285,068.24**. At payment time, buy pounds at $\min(S_T, 1.57)$, before including premium.

ExportCo: $d_1 = 0.2634591$ and $d_2 = 0.2034591$. Put price = **\$0.02674811 per pound**. For £30 million, premium = **\$802,443.22**. At receipt time, obtain $\max(S_T, 1.53)$ per pound, before including premium.

WHAT IS ACTUALLY GUARANTEED?

The call caps the transaction rate at 1.57; the put floors it at 1.53. The book's wording "less than" or "at least" concerns the exercise/transaction rate. The total effective cost or revenue includes the premium and its financing, so it is not the same number. Small software differences can come from day-count and rate conventions.

QUESTIONS 2.1-2.7 · SOURCE P. 67

37. Chapter 2 solutions: mechanics and margin

2.1 | Open interest versus volume

Open interest counts outstanding contracts at a moment; volume counts transactions over an interval. One long-short pair contributes one outstanding contract. Repeated trading can produce high volume with unchanged or falling open interest.

2.2 | Local versus FCM

In the chapter's terminology, a local trades for its own account. A futures commission merchant handles customer orders and earns commissions. The distinction concerns whose account is being served, not whether the economic purpose is hedging or speculation.

2.3 | Short 5,000 ounces of silver at 17.20

Initial margin \$4,000; maintenance \$3,000. Cushion is \$1,000. A short loses when price rises. A rise of $1,000/5,000 = \$0.20$ takes the balance exactly to maintenance; a rise **above \$0.20**, to a price **above \$17.40**, triggers a call under the strict "below maintenance" rule. Failure to meet it leads the broker to close the short by buying the same futures, with any residual debt still owed.

2.4 | Long crude oil across year-end

Total profit = $1,000 \times (90.50 - 88.30) = \$2,200$, realized in cash through daily settlement. Nonhedge annual amounts: 2015 = $1,000 \times (89.10 - 88.30) = \800 ; 2016 = $1,000 \times (90.50 - 89.10) = \$1,400$. As a properly identified business hedge, ordinary treatment is matched to the hedged item; if that transaction is recognized in 2016, the \$2,200 is matched there. As a corporate speculator, annual capital recognition applies in the book's framework. No tax bill is calculable without rates and other facts.

2.5 | Sell stop at 2 versus sell limit at 2

A sell stop at 2 is normally below the current market and activates after a fall to the trigger, often to exit a long. Execution can be below 2. A sell limit at 2 permits a sale only at 2 or higher, commonly placed above market to seek a favorable sale. It may not execute.

2.6 | Clearing-house versus broker margin

In the book's customer model, the broker allows a balance between initial and maintenance before demanding restoration to initial. Clearing members pay daily settlement losses and maintain the required initial margin for cleared positions; there is no analogous customer-style maintenance cushion in Hull's member model. New contracts can also add initial margin requirements.

2.7 | FX quote conventions

The chapter's currency futures are quoted in USD or US cents per foreign-currency unit. Spot/forward quotes for GBP, EUR, AUD and NZD use USD per foreign unit; other major currencies commonly use foreign units per USD. Convert reciprocals and cents before comparing. State these as the edition's convention, not an assumption about every available instrument.

QUESTIONS 2.8-2.14 · SOURCE PP. 67-68

38. Chapter 2 solutions: design and price triggers

2.8 | Delivery options held by the short

Additional flexibility over eligible assets, location or timing benefits the short. Buyers receive a less favorable obligation because the short selects what is cheapest or most convenient within the rules. Other things equal, these options **decrease the futures price** relative to an otherwise equivalent contract without them.

2.9 | Design a new contract

Specify underlying/grade, quantity, delivery location, listed months and permitted delivery dates, cash or physical settlement, quotation and tick, trading hours and last trading day, and applicable price/position limits. The contract should meet hedging needs and attract enough participation for liquidity. Contract size balances accessibility against per-contract costs.

2.10 | Why margin reduces default losses

Initial margin provides a buffer for adverse moves and liquidation. Daily settlement prevents unpaid losses from accumulating for months. Calls demand more cash as needed and closeout stops further exposure when funds fail to arrive. This reduces credit risk but does not guarantee that a discontinuous price move cannot exceed collateral.

2.11 | Two orange-juice contracts

Quantity = $2 \times 15,000 = 30,000$ pounds. Initial margin = \$12,000; maintenance = \$9,000. A \$3,000 loss uses the cushion, equivalent to $\$3,000/30,000 = \mathbf{\$0.10 = 10 \text{ cents per pound}}$. A fall below 150 cents from 160 triggers a call. To withdraw \$2,000, the balance must reach at least \$14,000 under unchanged requirements. Starting without other flows, that requires a rise of $2,000/30,000 = \mathbf{6.6667 \text{ cents}}$, to 166.6667 cents.

2.12 | Convergence during the delivery period

If futures exceeds spot enough to cover costs, buy the eligible asset, short futures and deliver. If futures is below spot, buying futures does not necessarily provide the asset immediately because delivery timing belongs to the short. Thus the reverse opportunity is not automatically riskless throughout the delivery period. At final matching settlement, convergence is enforced by the delivery/reference mechanism.

2.13 | MIT versus stop

A sell MIT typically activates after a favorable rise and can take profit on a long; a sell stop activates after an adverse fall and can cut exposure. The buy-side directions reverse. Both become market-type instructions after triggering; neither universally guarantees execution exactly at the trigger.

2.14 | Sell stop 20.30, limit 20.10

When the stop condition at 20.30 is met, the instruction becomes a sell limit at 20.10. It may sell at 20.10 or higher. If prices gap below 20.10 and do not recover, the order may remain unfilled. The 20.30 trigger and 20.10 minimum sale price perform different jobs.

QUESTIONS 2.15-2.22 · SOURCE P. 68

39. Chapter 2 solutions: clearing and participation

2.15 | A member adds 20 long contracts

The existing 100 contracts gain $100 \times (50,200 - 50,000) = \$20,000$. The 20 new contracts lose $20 \times (50,200 - 51,000) = \$16,000$. Net settlement gain is \$4,000. Additional initial margin is $20 \times \$2,000 = \$40,000$. Net additional cash required = $40,000 - 4,000 = \$36,000$, if settlement proceeds are available to meet the margin increase. Distinguish the \$40,000 gross margin requirement from the \$36,000 net funding.

2.16 | Why post-crisis rules raise collateral needs

More trades are centrally cleared, requiring initial and variation margin; more bilateral institutional trades also face margin requirements. Initial margin may be segregated and unavailable for reuse. Hence more eligible collateral and liquidity can be needed, even as unsecured exposure falls. Scope and exemptions matter.

2.17 | Swiss franc forward 1.1000, futures 0.9000

Forward convention here is CHF per USD, so 1.1000 means **1/1.1000 = \$0.9090909 per CHF**. Futures is already \$0.9000 per CHF. Someone selling CHF receives more USD under the quoted forward: a difference of about **\$0.0090909 per CHF**, before costs and timing effects.

2.18 | Instruct a broker to sell one July hogs contract

The broker checks account permissions and funds, routes the sell instruction, and obtains a match under the order conditions. Clearing records a short July position, margin is posted and daily settlement starts. A price fall credits the short; a rise debits it. To close, buy the same July contract. If retained, follow that product's final-settlement rules; do not infer actual live-animal delivery merely from the word hogs.

2.19 | Is speculation merely socially useless gambling?

Speculators take risks hedgers may wish to transfer and can contribute liquidity and price discovery. Hedgers need willing counterparties and may face lower trading costs when markets are active. Speculation can also amplify stress or support abusive conduct when controls fail. The economic benefits do not imply every speculative action is beneficial or profitable.

2.20 | Bilateral versus central clearing

Bilateral trades leave the original counterparties facing each other under a master agreement and collateral terms. A CCP accepts trades and becomes counterparty to each side, with margin and default procedures. Central clearing can facilitate multilateral netting but concentrates risk-management responsibilities in the CCP.

2.21 | Incomplete quality specification

Shorts would have an incentive to deliver the cheapest low-quality eligible goods, and disputes would arise about what is acceptable. Buyers would discount the contract or avoid it. Liquidity and usefulness for hedging would suffer. Precise quality standards and adjustment rules reduce this adverse-selection problem.

2.22 | Three possible changes in open interest

Both sides open: +1. Both close: -1. One opens and the other closes: 0. Every one-contract trade adds one to volume. A buyer is not necessarily opening a long; the buyer may be closing a short.

QUESTIONS 2.23-2.26 · SOURCE PP. 68-69

40. Chapter 2 solutions: business hedging

2.23 | Short live cattle across year-end

Quantity = 40,000 pounds. Total gain = $40,000 \times (1.2120 - 1.1830) = \text{\$1,160}$. At year-end: $40,000 \times (1.2120 - 1.1880) = \text{\$960 in 2015}$. Remaining gain: $40,000 \times (1.1880 - 1.1830) = \text{\$200 in 2016}$. Cash settles daily. Corporate speculation uses annual capital recognition under the exercise framework. A qualifying business hedge uses ordinary treatment and matches recognition to the hedged item; if the relevant sale is in 2016, match the \$1,160 there.

2.24 | Farmer will sell 120,000 pounds

Short $120,000/40,000 = 3$ **matching live-cattle futures contracts**. Falling futures prices produce gains that offset weaker sales proceeds. Advantages include greater budget certainty and protection against a price fall. Costs include giving up favorable price rises, margin funding, transaction costs and possible basis/quantity/timing mismatch. The farmer does not need to make exchange delivery: it can close futures and sell locally, leaving basis risk.

2.25 | A mine with output over the following year

Construction from July 2014 takes about six months, so output begins around January 2015 and continues through 2015. Forecast ounces in each sales interval, then short an appropriate ladder of 2015 futures maturities. For each 100 ounces to be hedged, use one contract, adjusting for rounding and uncertainty. Match maturities to expected sales; use later maturities or rolls where dates do not line up.

Do not short the whole expected annual output in a single nearby maturity merely because a contract is available. Staggering better matches the timing of production. Close positions as gold is sold, or deliver only if feasible under the contract. Update the hedge if production estimates change, avoid overhedging unproduced gold, and reserve cash for margin calls if gold prices rise.

2.26 | How CCPs work and why regulators use them

After acceptance, a CCP faces both parties, values exposures, collects margin, settles or collateralizes changes, and manages defaults using its financial resources and procedures. Advantages can include consistent risk management, transparency, multilateral netting and reduced unsecured bilateral chains. These can reduce contagion from a single counterparty's failure.

The benefits are conditional: a CCP concentrates risk, depends on sound margin/default management, and can impose large cash calls during stress. Several CCPs and excluded bilateral products can fragment netting. Requiring central clearing does not abolish market or systemic risk.

QUESTIONS 2.27-2.29 · SOURCE P. 69

41. Chapter 2 solutions: timing and year-end

2.27 | EUR 1 million: fall first, then finish at 1.3300

Both traders entered at 1.3000 USD/EUR. Ignoring daily-settlement financing, both make $1,000,000 \times (1.3300 - 1.3000) = \$30,000$. The futures trader pays losses during the early fall and receives gains later. With positive funding costs and the described path, the forward trader generally does better economically, because the forward avoids those early cash outflows. Exact difference requires the full path and interest/funding information; at zero financing rates, the nominal totals match.

2.28 | Count outstanding positions

Open interest is outstanding long-short pairs, not transactions per day. It usually declines as near-expiry contracts are closed or rolled. New sides = $600 + 800 = 1,400$; closing sides = $1,400 + 1,200 = 2,600$. Therefore change = $(1,400 - 2,600)/2 = -600$ contracts. Independently, net change in long positions is 600 opening buyers minus 1,200 closing sellers = -600.

2.29 | Short orange juice across three calendar years

One contract covers 15,000 pounds. Entry is \$1.20/lb; year-end prices are \$1.40 and \$1.10; exit is \$1.25. The short's daily P&L is quantity $\times$ (previous price - new price).

Period	Calculation	Gain or loss
2014	$15,000 \times (1.20 - 1.40)$	-\$3,000
2015	$15,000 \times (1.40 - 1.10)$	+\$4,500
2016 to exit	$15,000 \times (1.10 - 1.25)$	-\$2,250
Total	$15,000 \times (1.20 - 1.25)$	-\$750

Cash: all gains and losses are settled daily, even for a hedge. **Nonhedge accounting and speculative tax treatment:** recognize the annual changes shown, with the tax classification appropriate to the taxpayer under the chapter framework. **Qualifying hedge:** accounting recognition follows the hedged item and tax treatment is ordinarily matched business-hedge treatment. If the hedged transaction is recognized in 2016, the combined \$750 loss is matched then in the simplified exercise model.

WHY THIS QUESTION MATTERS

A cash margin payment in 2014 can coexist with hedge-related income recognition in 2016. "When cash is paid" and "which year recognizes the result" are different answers. The exercise does not state enough to compute a company's actual tax liability.

QUESTIONS 2.30-2.34 · SOURCE PP. 69-70

42. Chapter 2 solutions: margin, carry and credit

2.30 | Short wheat at 750 cents

Quantity 5,000 bushels; initial margin \$3,000 and maintenance \$2,000. A short loses after a rise. The \$1,000 cushion equals \$0.20 or 20 cents per bushel. A price **above 770 cents** triggers a call under a strict below-maintenance rule. To withdraw \$1,500 from an otherwise unchanged account, gain \$1,500: price must fall \$0.30 or 30 cents to **720 cents or below**.

2.31 | June oil at 80, December at 86

Buy June futures and sell December futures, arranging to take delivery in June, carry the oil and deliver in December. Under the question's zero-storage assumption and a simple six-month 5% annual financing calculation, \$80 borrowed in June grows to $80 \times (1 + 0.05 \times 0.5) = \mathbf{\$82}$. Receive \$86 in December, giving approximately **\$4 per barrel**, or \$4,000 for a 1,000-barrel exposure.

With continuous compounding the carrying amount is $80 \exp(0.05 \times 0.5) = \82.0252 and the difference is \$3.9748. The introductory answer ignores interest effects of intermediate futures settlement and assumes feasible matching delivery and funding. The \$4 should not be presented as an unconditional exact realized profit after all margin-financing effects.

2.32 | Long forward plus matching put

The total payoff is $S_T - K + \max(K - S_T, 0) = \mathbf{\max(S_T - K, 0)}$, the payoff of a long call. If K equals the fair forward price, a zero-value forward plus the fairly priced put replicates the call with matching initial value. For an arbitrary off-market K , account for the forward's initial value as well.

2.33 | Bilateral collateral versus one CCP

(a) Under one-way bilateral agreements, post \$15 million to B and \$25 million to C; receive nothing on the \$20 million claim against A. Total posted = **\$40 million**. (b) Through the same CCP, total market value is $20 - 15 - 25 = -\$20$ million. Add \$10 million initial margin: total posted = **\$30 million**, assuming the stated netting and no other thresholds.

2.34 | A \$10 million claim with matching cash collateral

Current unsecured exposure is approximately zero if the collateral is available and enforceable. Residual risk remains during the interval after default until closeout. If the derivative value rises to \$13 million while collateral stays at \$10 million, the new gap is \$3 million. An offsetting hedge can lose cash while the defaulted counterparty fails to pay the derivative gain. Collateral reduces exposure; it does not fix future exposure permanently.

QUESTION 2.35(A)-(B) · SOURCE P. 70

43. Chapter 2 solution: estimating a margin

DATA STATUS

This question depends on a separate crude-oil price history, not included in the uploaded PDF. The author's linked data pages returned an unavailable/error page. The historical standard deviation and historical failure count are therefore not claimed here. The complete method below is ready to use with the assigned file.

What the question is asking

Use daily futures price changes to estimate how much the market might move against a position before additional collateral arrives. This is a probability model for potential loss over a short horizon, not an option-pricing calculation.

(a) Estimate daily dispersion

Sort observations from oldest to newest. Define $x_t = F_t - F_{t-1}$ in dollars per barrel. Estimate the daily standard deviation s using the sample standard deviation of these changes. A zero-mean maximum-likelihood convention instead uses the square root of the mean squared change; specify the estimator because results differ slightly.

CALCULATION

Sample $s = \sqrt{\text{sum}((x_t - \text{mean}(x))^2) / (n - 1)}$

One-sided 99% adverse-move threshold = $2.32635 \times s$

If "movement not exceeded" means the absolute movement in either direction with 99% confidence, use **2.57583s**, since 0.5% lies in each tail. For protecting one fixed long or short position against a 1% adverse tail, use **2.32635s**. State which interpretation you use.

(b) Extend to two days

Under independent, constant-variance, zero-mean daily normal changes, the two-day variance is $2s^2$, so standard deviation is $\sqrt{2} s$. For a long position, loss is $-Q(F_{t+2} - F_t)$. Therefore:

CALCULATION

Maintenance margin $M = Q \times 2.32635 \times \sqrt{2} \times s$

Initial margin $I = M / 0.75$

The multiplier Q is 1,000 barrels for the chapter's crude-oil example. Use the assigned contract's actual multiplier if different. Independence is an additional assumption: normal daily marginals alone do not establish the $\sqrt{2}$ rule.

Illustration, not the missing historical answer

If s were \$1.50 per barrel, the one-day adverse threshold would be \$3.4895 per barrel. Maintenance would be $1,000 \times 2.32635 \times \sqrt{2} \times 1.50 =$ about **\$4,934.93**, and initial margin about **\$6,579.91**. A two-day loss beyond that maintenance amount is a 1% event under this model.

QUESTION 2.35(C) · REPRODUCIBLE PROCEDURE

44. Chapter 2 solution: backtesting and code

Compute every observed two-day change $y_t = F_{t+2} - F_t$. For a long, a balance initially at maintenance would be wiped out when $Qy_t < -M$. Count these events and divide by the number of two-day windows. Overlapping windows are dependent; do not treat the resulting observations as independent statistical trials.

Because part (c) supplies initial margin at $M/0.75$, also simulate the margin account: start at I , post losses/gains daily, withdraw excess above I , and call for restoration when the balance drops below M . In the book's timing, calls issued at day-end are paid by the next day-end. Distinguish a negative balance before that payment from a simple two-day return exceedance.

Save the assigned data in chronological order as a CSV with a numeric "price" column in dollars per barrel. The following uses only the Python standard library. It prints both diagnostics and states the convention explicitly.

```
import csv, math, statistics
from statistics import NormalDist
with open("crude_oil.csv", newline="") as f:
    prices = [float(r["price"]) for r in csv.DictReader(f)]
assert len(prices) >= 3
Q = 1000
x = [b-a for a,b in zip(prices, prices[1:])]
s = statistics.stdev(x)
z = NormalDist().inv_cdf(0.99)
M = Q * z * math.sqrt(2) * s
I = M / 0.75
moves2 = [prices[t+2]-prices[t]
           for t in range(len(prices)-2)]
exceed = sum(Q*d < -M for d in moves2)
print("Daily s; one-sided 99%:", s, z*s)
print("Maintenance; initial:", M, I)
print("Two-day exceedances:", exceed, len(moves2),
      exceed/len(moves2))
balance, due = I, 0.0
calls, negative = 0, 0
for d in x:
    balance += Q*d
    negative += (balance < 0) # before today's top-up
    balance += due           # previous day's call
    due = 0.0
    balance = min(balance, I) # withdraw excess
    if balance < M:
        due = I - balance    # due next day-end
        calls += 1
print("Calls; negative pre-top-up observations:",
      calls, negative)
```

Interpretation: materially more adverse events than the nominal model predicts can suggest heavy tails, changing volatility, dependence or data/roll issues. Inspect the sample and conventions before blaming one cause. A synthetic example cannot establish the actual historical failure rate.

QUICK REFERENCE

45. Formula sheet: positions and hedges

Quantity	Formula or rule
Long forward payoff	$Q(S_T - K)$
Short forward payoff	$Q(K - S_T)$
Long futures P&L	$N \times \text{multiplier} \times (\text{exit price} - \text{entry price})$
Short futures P&L	Negative of the matching long result
Long call profit	$Q[\max(S_T - K, 0) - c]$
Long put profit	$Q[\max(K - S_T, 0) - p]$
Short call/put profit	Negative of the matching buyer profit
Call break-even	$K + c$, ignoring financing
Put break-even	$K - p$, ignoring financing
Protective put gross value	$Q \max(S_T, K)$
Forward + put payoff	$Q \max(S_T - K, 0)$
Call - put payoff	$Q(S_T - K)$
Call-spread payoff	$Q[\max(S_T - K_1, 0) - \max(S_T - K_2, 0)]$

The cash-flow checklist

- Is the quoted premium per share, per underlying unit or per contract?
- Is the price in dollars, cents, index points or a reciprocal currency quote?
- Are you calculating payoff, profit, balance or a required deposit?
- Are cash flows being compared at the same date?
- If hedging, have you combined the derivative with the original exposure?

MAXIMUM-LOSS LANGUAGE

For a bought option, state “loss of the standalone option premium.” For a stock put writer, state the nonnegative-stock assumption. For futures, say margin does not cap losses. Avoid saying every derivative has unlimited loss or every option has premium-limited loss.

QUICK REFERENCE

46. Formula sheet: margin, quotes and arbitrage

Task	Method
Margin buffer	Initial margin - maintenance margin
Price move to maintenance	Buffer / total underlying quantity
Top-up after a call	Initial margin - current balance
Withdrawal under textbook rule	Balance above required initial margin
Tick value	Contract multiplier $\times$ minimum quote increment, with unit conversion
Open interest change	(Opening sides - closing sides)/2
Reciprocal FX quote	1 / original rate; also reverse bid and ask
Contract count	Exposure quantity / units per contract, subject to rounding and mismatch
No-income one-year forward	$S_0(1 + r)$, under simple annual-rate assumptions
Two-day margin estimate	$Q \times 2.32635 \times \sqrt{2} \times \text{daily standard deviation}$, for a 1% one-sided normal tail

A general margin-call derivation

Let account balance before the move be B and maintenance requirement be M . A long experiences loss Qd after a price fall d . The call condition is $B - Qd < M$, or $d > (B - M)/Q$. For a short, replace the fall with a rise. Use B , not automatically initial margin, if the account already contains prior gains/losses.

A general arbitrage checklist

Identify the same final asset or cash flow in two ways. Lock the trades at executable prices. Include borrowing/lending, storage, income, FX conversion and fees. Verify that delivery dates and quantities match. Show initial net cash flow and terminal cash flow. Finally ask whether any unhedged funding, delivery, short-sale or credit assumption remains.

EQUALITY AT THE THRESHOLD

Hull's margin rule says below maintenance. A move that puts the account exactly at maintenance reaches the boundary; a move beyond it triggers the call. State this explicitly when an exercise asks what price change leads to a call.

DEFINITIONS FOR ORAL QUESTIONS

47. Glossary: contracts and valuation

Term	Clear definition
Underlying	The variable or asset on which a derivative depends.
Forward	A bilateral future transaction commitment at an agreed delivery price.
Futures	A standardized, cleared exposure with regular settlement of gains and losses.
Call / put	A right to buy / a right to sell under specified exercise terms.
Holder / writer	The owner of an option right / the seller accepting its corresponding obligation.
Strike	Price used when an option is exercised.
Premium	Price paid to obtain an option.
Exercise / assignment	Holder uses a right / the obligation is allocated to a writer under the rules.
American / European	Exercise permitted during the allowed life / only at expiration.
Moneyness	Whether immediate exercise would have positive value; it ignores premium paid.
Payoff / profit	Terminal contractual result / result after relevant acquisition costs and receipts.
Spot / forward price	Immediate-delivery quote / agreed-price quote for a new future-delivery transaction.
Market value	Current value of a contract's remaining rights and obligations.
Notional	Reference size; distinct from current value and maximum possible loss.
Zero-coupon bond	Instrument with a specified maturity payment and no intermediate coupon.
Swap	Agreement to exchange cash flows under specified rules on a series of dates.
Replication	Combining instruments to produce the same specified cash flows as another.
Leverage	Exposure large relative to the capital initially committed.

DEFINITIONS FOR ORAL QUESTIONS

48. Glossary: market plumbing and risk

Term	Clear definition
Bid / offer	Dealer's buying / selling quote; customers normally sell at bid and buy at offer.
Market maker	Participant willing to quote both sides and take positions as principal.
Liquidity	Ability to transact promptly without a large price impact.
Clearing house / CCP	Institution interposed between cleared counterparties to manage obligations.
FCM / local	Customer-serving futures intermediary / own-account trader in floor terminology.
Initial / maintenance margin	Required opening collateral / customer threshold for replenishment.
Marking to market	Revaluing using current prices; for futures, accompanied by cash settlement.
Variation margin	A term used for price-change settlement or collateral flows, depending on context.
CSA	Credit support annex specifying bilateral collateral arrangements.
Haircut	Discount applied to collateral value for margin purposes.
Netting	Combining eligible obligations under enforceable arrangements.
Guaranty/default fund	Resources contributed to support management of clearing-member default.
Open interest / volume	Outstanding contracts / contracts traded over an interval.
Basis	Difference between matched spot and futures prices, often $S - F$.
Calendar spread / roll	Opposite positions in different maturities / close one maturity and open another.
Systemic risk	Risk of disruption spreading through the financial system.
Funding liquidity risk	Inability to obtain cash when payment or collateral is due.
Basis risk	Risk that the hedge and the actual exposure do not offset exactly.
Front running	Misusing knowledge of a customer order to trade first for oneself.

INTERNSHIP PRACTICE

49. Oral rehearsal: explain, then defend

A | Why can a zero-cost forward be valuable later?

Zero cost initially means its original terms were fair at inception. Market conditions change. A fixed old delivery price can become favorable or unfavorable relative to new contracts. Distinguish delivery price from contract value.

B | Why does an option seller accept the obligation?

The seller receives a premium and may hedge or hold a portfolio for which the exposure is useful. Receiving premium is compensation for an obligation, not guaranteed income. Ask which risk the writer retains.

C | How can both parties benefit from a zero-sum contract?

Cash gains and losses offset, but parties value risk differently. A farmer can value revenue certainty; another participant can willingly take that risk. Economic usefulness includes risk transfer, not just ex-post trading profit.

D | Why not always choose options instead of forwards?

Options preserve favorable outcomes but cost a premium. A business may prefer the lower initial purchase outlay and price certainty of a forward. The right choice depends on objectives, exposure uncertainty, cost and funding.

E | Why does central clearing require margin from both sides?

Either long or short can lose after the next price move. A matched clearing book does not remove the risk that one member fails to make the required payment while another must still be paid.

F | Can a perfect hedge still be dangerous?

A payoff hedge can leave timing risk: margin must be paid before gains on the underlying turn into cash. Operational, legal and counterparty risks can remain as well. Describe the cash-flow dates, not only final payoffs.

G | Why does a short delivery option lower the futures price?

The short can choose a cheaper permitted way to fulfill the contract. The long receives a less valuable promise than under a fully fixed delivery requirement. The agreed price adjusts for that flexibility.

H | Does a stop-loss guarantee a maximum loss?

No. It triggers an execution instruction. Market gaps or unavailable liquidity can yield a worse price. A stop-limit can protect execution price but fail to execute. An option supplies a contractual payoff over its specified horizon.

I | What is the first thing to check in a numerical question?

Units, multiplier, direction and dates. A correct formula with cents mistaken for dollars or contracts mistaken for shares can be wrong by factors of 100 or more.

ORIGINAL EXERCISES WITH ANSWERS

50. Extra practice: transfer the method

P1 | A currency invoice

A firm owes \$80,000 in three months and locks 10.25 MAD/USD. Spot finishes at 10.70. Calculate the hedge advantage and final effective cost. **Answer:** advantage = $80,000 \times 0.45 = 36,000$ MAD; cost = $80,000 \times 10.25 = 820,000$ MAD. Verify 856,000 spot cost minus 36,000 gain.

P2 | A put with an exercise loss but a trading profit for the writer

Write a put at strike 70 for premium 6. Final stock is 67. **Answer:** holder exercises for 3; writer profit = $6 - 3 = 3$ per share. Assignment does not automatically imply a net losing trade.

P3 | A margin account already contains losses

A long account balance is \$7,400; initial margin is \$8,000 and maintenance is \$6,000. Total quantity is 2,000 units. What fall reaches maintenance, and what deposit follows a fall of \$0.90? **Answer:** boundary fall = $1,400/2,000 = \$0.70$. A \$0.90 fall loses \$1,800, leaving \$5,600. Top-up = \$2,400.

P4 | Closing versus changing maturity

You are long five June contracts and sell five September contracts. Are you flat? **Answer:** no. You hold a calendar spread. You need to sell five June contracts to close the June position; the September short remains unless separately closed.

P5 | Why a bigger notional does not determine credit exposure

A \$20 million-notional derivative is worth \$100,000 to you and is fully collateralized. Is your current unsecured claim \$20 million? **Answer:** no; notional is a reference scale. Current unsecured exposure can be near zero, while future closeout exposure can still develop.

P6 | Derive a floor instead of remembering it

Own 200 shares bought at 50 and buy 200 puts at strike 46 for premium 2. **Answer:** initial total cost = $200 \times 52 = \$10,400$. Gross terminal floor = $200 \times 46 = \$9,200$. Maximum loss on the combined investment is \$1,200, ignoring financing. Netting out the premium from a holding-value graph is equivalent if applied consistently.

P7 | Count open interest

During a session 700 sides open and 1,100 sides close. What is the change? **Answer:** $(700 - 1,100)/2 = -200$ contracts. Counting sides is not the same as counting trades.

P8 | A futures winner with a funding problem

A position ends with a positive cumulative P&L but has a large early drawdown. Can the trader fail? **Answer:** yes, if interim settlement payments cannot be funded. Final profitability and the ability to survive the path are different.

AUDIT YOUR PREPARATION

51. Coverage checklist and source notes

Book section	Teaching section in this guide
1.1-1.2 · Venues, electronic markets, market size	2; business cases in 11
1.3 · Forwards, payoffs, spot/forward relation	3-5
1.4 · Futures	5, then 12-25
1.5 · Options and option quotes	6-8
1.6-1.7 · Trader types and hedging	9; currency cases in 4
1.8 · Speculation	10
1.9-1.10 · Arbitrage and dangers	5 and 11
2.1-2.3 · Background, specification, convergence	12-14
2.4 · Customer and member margin accounts	15-17
2.5 · Bilateral/central clearing, collateral	18
2.6 · Quotes, volume, interest and patterns	19-20
2.7-2.8 · Delivery, traders and orders	21-22
2.9-2.11 · Regulation, accounting/tax, forwards	23-25
Questions 1.1-1.41	27-36; 1.41 is grouped with payoff combinations in 35
Questions 2.1-2.35	37-44; 2.35 includes an explicit data limitation

Book examples and snapshots

Coverage includes Table 1.1 currency quotes; Tables 1.2-1.3 Google options; ImportCo/ExportCo; protective puts; sterling futures speculation; stock-versus-call speculation; the dual-listed stock and \$60 carry example; corn trade creation; contract design examples; the complete Table 2.1 gold path; Table 2.2 quotation interpretation; clearing-member netting; the corn accounting example; and the £1 million forward/futures timing comparison. All six numbered Business Snapshots are discussed in teaching sections 11, 12 and 18. The further-reading bibliographies are not additional examinable theories.

Sources and calculation provenance

Primary source: John C. Hull, *Options, Futures, and Other Derivatives*, 9th Global Edition, Pearson, ISBN 978-1-292-21289-0; user-supplied PDF, printed pp. 23-70. All exercise answers here are independently explained and are not represented as the publisher's official solutions.

Question 2.35 external source named by the book: www-2.rotman.utoronto.ca/~hull/data/. Retrieval was unsuccessful; a dataset-specific result requires the assigned price file. Question 1.40 uses a direct numerical implementation of the European FX option formula, with all inputs shown. Charts are newly drawn; extra examples are labeled.